

3

Oscillators

Harmonic oscillators are of enormous importance for physics because most of condensed-matter physics and quantum electrodynamics centre on weakly perturbed harmonic oscillators. The reason harmonic oscillators are so common is simple. The points of equilibrium of a particle that moves in a potential $V(x)$ are points at which the force $-dV/dx$ vanishes. When we place the origin of x at such a point, the Maclaurin expansion of V becomes $V(x) = \text{constant} + \frac{1}{2}V''x^2 + O(x^3)$, and the force on the particle becomes $F = -V''x + O(x^2)$. Consequently, for sufficiently small excursions from the point of equilibrium, the particle's motion will be well approximated by a harmonic oscillator.

Besides providing the background to a great many branches of physics, our analysis of a harmonic oscillator will introduce a technique that will prove invaluable on several subsequent occasions.

3.1 Stationary states of a harmonic oscillator

We can build a harmonic oscillator by placing a particle in a potential that increases quadratically with distance from the origin. Hence an appropriate Hamiltonian is given by equation (2.51) with $V \propto x^2$.¹ For later convenience we choose the constant of proportionality such that H

¹ In the last chapter we distinguished the position and momentum operators from their eigenvalues with hats. Henceforth we drop the hats; the distinction between operator and eigenvalue should be clear from the context.

becomes

$$H = \frac{1}{2m} \{p^2 + (m\omega x)^2\}. \quad (3.1)$$

In §2.2 we saw that the dynamical evolution of a system follows immediately once we know the eigenvalues and eigenkets of H . So we now determine these quantities for the Hamiltonian (3.1).

We next introduce the dimensionless operator

$$A = \frac{m\omega x + ip}{\sqrt{2m\hbar\omega}}. \quad (3.2a)$$

This operator isn't Hermitian. Bearing in mind that x and p are Hermitian, from the rules in Table 2.1 we see that its adjoint is

$$A^\dagger = \frac{m\omega x - ip}{\sqrt{2m\hbar\omega}}. \quad (3.2b)$$

The product $A^\dagger A$ is

$$\begin{aligned} A^\dagger A &= \frac{1}{2m\hbar\omega} (m\omega x - ip)(m\omega x + ip) \\ &= \frac{1}{2m\hbar\omega} \{ (m\omega x)^2 + im\omega[x, p] + p^2 \} = \frac{H}{\hbar\omega} - \frac{1}{2}, \end{aligned} \quad (3.3)$$

where we have used the canonical commutation relation (2.54). This equation can be rewritten $H/(\hbar\omega) = A^\dagger A + \frac{1}{2}$, so A is rather nearly the square root of the dimensionless Hamiltonian $H/\hbar\omega$. If we calculate $AA^\dagger$ in the same way, the only thing that changes is the sign in front of the commutator $[x, p]$, so we have

$$AA^\dagger = \frac{H}{\hbar\omega} + \frac{1}{2}. \quad (3.4)$$

Subtracting equation (3.4) from equation (3.3) we find that

$$[A^\dagger, A] = -1. \quad (3.5)$$

We will find it useful to have evaluated the commutator of $A^\dagger$ with the Hamiltonian. Since from equation (3.3) $H = \hbar\omega(A^\dagger A + \frac{1}{2})$, we can write

$$[A^\dagger, H] = \hbar\omega[A^\dagger, A^\dagger A] = \hbar\omega A^\dagger[A^\dagger, A] = -\hbar\omega A^\dagger, \quad (3.6)$$

where we have exploited the rules of equations (2.22).

We now multiply both sides of the defining relation of $|E_n\rangle$, namely $H|E_n\rangle = E_n|E_n\rangle$, by $A^\dagger$:

$$A^\dagger E_n|E_n\rangle = A^\dagger H|E_n\rangle = (HA^\dagger + [A^\dagger, H])|E_n\rangle = (H - \hbar\omega)A^\dagger|E_n\rangle. \quad (3.7)$$

A slight rearrangement of this equation yields

$$H(A^\dagger|E_n\rangle) = (E_n + \hbar\omega)(A^\dagger|E_n\rangle). \quad (3.8)$$

Provided $|b\rangle = A^\dagger|E_n\rangle$ has non-zero length-squared, this shows that $|b\rangle$ is an eigenket of H with eigenvalue $E_n + \hbar\omega$. The length-square of $|b\rangle$ is

$$|A^\dagger|E_n\rangle|^2 = \langle E_n|AA^\dagger|E_n\rangle = \langle E_n|\left(\frac{H}{\hbar\omega} + \frac{1}{2}\right)|E_n\rangle = \frac{E_n}{\hbar\omega} + \frac{1}{2}. \quad (3.9)$$

Now squeezing H between $\langle E_n|$ and $|E_n\rangle$ we find with (3.1) that

$$E_n = \langle E_n|H|E_n\rangle = \frac{1}{2m\omega}(|p|E_n\rangle|^2 + m^2\omega^2|x|E_n\rangle|^2) \geq 0. \quad (3.10)$$

Thus the energy eigenvalues are non-negative, so $|A^\dagger|E_n\rangle|^2 > 0$ and by repeated application of $A^\dagger$ we can construct an infinite series of eigenstates with energy $E_n + k\hbar\omega$ for $k = 0, 1, \dots$

Similarly, we can show that provided $A|E_n\rangle$ has non-zero length-squared, it is an eigenket of H for energy $E_n - \hbar\omega$. Since we know that all eigenvalues are non-negative, for some energy E_0 , $A|E_0\rangle$ must vanish. Equating to zero the length-squared of this vector we obtain an equation for E_0 :

$$0 = |A|E_0\rangle|^2 = \langle E_0|\left(\frac{H}{\hbar\omega} - \frac{1}{2}\right)|E_0\rangle = \frac{E_0}{\hbar\omega} - \frac{1}{2}. \quad (3.11)$$

So $E_0 = \frac{1}{2}\hbar\omega$ and we have established that the eigenvalues of H are

$$\hbar\omega \times \left(\frac{1}{2}, \frac{3}{2}, \dots, \frac{2r+1}{2}, \dots\right) \quad \text{that is} \quad E_r = \left(r + \frac{1}{2}\right)\hbar\omega. \quad (3.12)$$

The operators $A^\dagger$ and A with which we have obtained these important results are respectively called **creation** and **annihilation operators** because the first creates an excitation of the oscillator, and the second destroys one. In quantum field theory particles are interpreted as excitations of the vacuum and each particle species is associated with creation and annihilation operators that create and destroy particles of the given species. A and $A^\dagger$ are also called **ladder operators**.

We now examine the eigenkets of H . Let $|r\rangle$ denote the state of energy $(r + \frac{1}{2})\hbar\omega$. In this notation the state of lowest energy, or **ground state**, is $|0\rangle$ and its defining equation is $A|0\rangle = 0$. From equation (3.2a) this equation reads

$$0 = A|0\rangle = \frac{m\omega x|0\rangle + ip|0\rangle}{\sqrt{2m\hbar\omega}}. \quad (3.13)$$

We now go to the position representation by multiplying through by $\langle x|$. With equations (2.48) and (2.49) we find that the equation becomes

$$\frac{1}{\sqrt{2m\hbar\omega}}\left(m\omega x + \hbar\frac{\partial}{\partial x}\right)\langle x|0\rangle = 0. \quad (3.14)$$

This is a linear, first-order differential equation. Its integrating factor is $\exp(m\omega x^2/2\hbar)$, so the correctly normalised wavefunction is

$$\langle x|0\rangle = \frac{1}{(2\pi\ell^2)^{1/4}}e^{-x^2/4\ell^2}, \quad \text{where} \quad \ell = \sqrt{\frac{\hbar}{2m\omega}}. \quad (3.15)$$

Notice that this solution is unique, so the ground state is **non-degenerate**. It is a Gaussian function, so the probability distribution $P(x) = |\langle x|0\rangle|^2$ for the position of the particle that forms the oscillator is also a Gaussian: its dispersion is ℓ .

From equations (2.64) and (2.65) we see that the momentum distribution of the wavefunction (3.15) is

$$P(p) = |\langle p|0\rangle|^2 \propto e^{-2\ell^2 p^2/\hbar^2}, \quad (3.16)$$

which is a Gaussian with dispersion $\sigma_p = \hbar/2\ell$. By inserting $x = \ell$ and $p = \sigma_p$ in the Hamiltonian (3.1) we obtain estimates of the typical kinetic and potential energies of the particle when it's in its ground state. We find that both energies are $\sim \frac{1}{4}\hbar\omega$. In fact one can straightforwardly show that $H(\ell, \sigma_p)$ is minimised subject to the constraint $\ell\sigma_p \geq \hbar/2$ when ℓ and σ_p take the values that we have derived for the ground state (Problem 3.4). In other words, in its ground state the particle is as stationary and as close to the origin as the uncertainty principle permits; there is a conflict between the advantage energetically of being near the origin, and the energetic penalty that the uncertainty principle exacts for having a well-defined position.

Every system that has a confining potential exhibits an analogous **zero-point motion**. The energy tied up in this motion is called **zero-point energy**. Zero-point motion is probably the single most important prediction of quantum mechanics, for the material world is at every level profoundly influenced by this phenomenon.

We obtain the wavefunctions of excited states by applying powers of the differential operator $A^\dagger$ to $\langle x|0\rangle$. Equation (3.9) enables us to find the normalisation constant α in the equation $|n+1\rangle = \alpha A^\dagger|n\rangle$; it implies that $\alpha^2 = n+1$. The generalisation of equation (3.11) enables us to determine the number β in the equation $|n-1\rangle = \beta A|n\rangle$, and we have finally

$$|n+1\rangle = \frac{1}{\sqrt{n+1}} A^\dagger|n\rangle \quad ; \quad |n-1\rangle = \frac{1}{\sqrt{n}} A|n\rangle. \quad (3.17)$$

It is useful to remember that the normalisation constant is always the square root of the largest value of n appearing in the equation. As a specific example

$$\begin{aligned} \langle x|1\rangle &= \frac{1}{\sqrt{2m\hbar\omega}} \left(m\omega x - \hbar \frac{\partial}{\partial x} \right) \langle x|0\rangle = \left(\frac{x}{2\ell} - \ell \frac{\partial}{\partial x} \right) \langle x|0\rangle \\ &= \frac{1}{(2\pi\ell^2)^{1/4}} \frac{x}{\ell} e^{-x^2/4\ell^2}. \end{aligned} \quad (3.18)$$

Whereas the ground-state wavefunction is an even function of x , the wavefunction of the first excited state is an odd function because $A^\dagger$ is odd in x . Wavefunctions that are even in x are said to be of **even parity**, while those that are odd functions have **odd parity**. It is clear that this pattern will be repeated as we apply further powers of $A^\dagger$ to generate the

other states of well-defined energy, so $\langle x|n\rangle$ is even parity if n is even, and odd parity otherwise.

Notice that the operator $N = A^\dagger A$ is Hermitian. By equations (3.17) $N|n\rangle = n|n\rangle$, so its eigenvalue tells you the number of excitations the oscillator has. Hence N is called the **number operator**.

Let's use these results to find the mean-square displacement $\langle n|x^2|n\rangle$ when the oscillator is in its n^{th} excited state. Adding equations (3.2) we express x as a linear combination of A and $A^\dagger$

$$x = \sqrt{\frac{\hbar}{2m\omega}}(A + A^\dagger) = \ell(A + A^\dagger), \quad (3.19)$$

where ℓ is defined by (3.15), so

$$\langle n|x^2|n\rangle = \ell^2 \langle n|(A + A^\dagger)^2|n\rangle. \quad (3.20)$$

When we multiply out the bracket on the right, the only terms that contribute are the ones that involve equal numbers of A s and $A^\dagger$ s. Thus

$$\langle n|x^2|n\rangle = \ell^2 \langle n|(AA^\dagger + A^\dagger A)|n\rangle = \ell^2(2n + 1) = \ell^2 \frac{2E_n}{\hbar\omega}, \quad (3.21)$$

where we have used equations (3.17) and (3.12). If we use equation (3.15) to eliminate ℓ , we obtain a formula that is valid in classical mechanics (Problem 2.23).

3.2 Dynamics of oscillators

By equations (2.29) and (3.12), the n^{th} excited state of the harmonic oscillator evolves in time according to

$$|n, t\rangle = e^{-i(n+1/2)\omega t}|n, 0\rangle. \quad (3.22)$$

Consequently, no state oscillates at the oscillator's classical frequency ω . How do we reconcile this result with classical physics?

We have seen that we make the link from quantum to classical physics by considering the expectation values of observables – if classical physics applies, the measured value of any observable will lie close to the expectation value, so the latter provides an accurate description of what's happening. Equation (2.35) tells us that when a system is in an energy eigenstate, the expectation value of any time-independent observable Q cannot depend on time. Equation (3.22) enables us to obtain this result from a different point of view by showing that when we form the expectation value $\langle Q \rangle = \langle \psi|Q|\psi \rangle$, the factor $e^{-iE_n t/\hbar}$ in the ket $|\psi, t\rangle = e^{-iE_n t/\hbar}|E_n\rangle$ cancels on the corresponding factor in $\langle \psi, t|$. Hence energy eigenstates are incapable of motion.² The system is capable of motion only if there are

² If we consider that t is the variable canonically conjugate to energy, this fact becomes a manifestation of the uncertainty principle.

non-negligible amplitudes to measure more than one possible energy, or, equivalently, if none of the a_i in the sum (2.32) has near unit modulus.

Consideration of the motion of a harmonic oscillator will make this general point clearer. If the oscillator's state is written

$$|\psi, t\rangle = \sum a_j e^{-iE_j t/\hbar} |j\rangle, \quad (3.23)$$

then the expectation value of x is

$$\langle x \rangle = \sum_{jk} a_k^* a_j e^{i(E_k - E_j)t/\hbar} \langle k|x|j \rangle = \sum_{jk} a_k^* a_j e^{i(k-j)\omega t} \langle k|x|j \rangle. \quad (3.24)$$

We simplify this expression by using equation (3.19) to replace x with $\ell(A + A^\dagger)$ and then using (3.17) to evaluate the matrix elements of A and $A^\dagger$:

$$\begin{aligned} \langle x \rangle &= \ell \sum_{jk} a_k^* a_j e^{i(k-j)\omega t} \langle k|(A + A^\dagger)|j \rangle \\ &= \ell \sum_{jk} a_k^* a_j e^{i(k-j)\omega t} (\sqrt{j} \langle k|j-1 \rangle + \sqrt{j+1} \langle k|j+1 \rangle). \end{aligned} \quad (3.25)$$

Since $\langle k|j-1 \rangle$ vanishes unless $k = j-1$, it's now easy to perform the sum over k , leaving two terms to be summed over j . On account of the factor $\sqrt{j}$ we can restrict the first of these sums to $j > 0$, and in the second sum we replace j by $j' = j+1$ and then replace the symbol j' by j so we can combine the two sums. After these operations we have

$$\begin{aligned} \langle x \rangle &= \ell \sum_{j=1} \sqrt{j} (a_{j-1}^* a_j e^{-i\omega t} + a_j^* a_{j-1} e^{i\omega t}) \\ &= \sum_j X_j \cos(\omega t + \phi_j), \end{aligned} \quad (3.26a)$$

where the real numbers X_j and ϕ_j are defined by

$$2\sqrt{j}\ell a_j^* a_{j-1} = X_j e^{i\phi_j}. \quad (3.26b)$$

Thus $\langle x \rangle$ oscillates sinusoidally at the classical frequency ω regardless of the amplitudes a_j . Thus we have recovered the classical result that the frequency at which a harmonic oscillator oscillates is independent of amplitude and equal to $\sqrt{k/m}$, where k is the oscillator's spring constant.

In the classical regime, the only non-negligible amplitudes a_j have indices j that cluster around some large number n . Consequently, a measurement of the energy is guaranteed to yield a value that lies close to $E = E_n$, and from equation (3.21) it follows that the mean value of x^2 will lie close to $\overline{x^2} = 2\ell^2 E_n / (\hbar\omega)$. Classically, the time average of x^2 is proportional to the average potential energy, which is just half the total energy. Hence, averaging the Hamiltonian (3.1) we conclude that classically $\overline{x^2} = E / (m\omega^2)$, in precise agreement with the quantum-mechanical result. The correspondence principle requires the classical and quantum-mechanical values of $\overline{x^2}$ to agree for large n . That they agree even for small n is a coincidence.

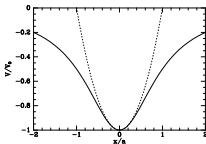


Figure 3.1 The potential energy $V(x)$ of an anharmonic oscillator (full curve) and $V(x)$ for the harmonic oscillator obtained by restricting the potential to the first two terms in its Maclaurin expansion (dashed curve).

3.2.1 Anharmonic oscillators

The Taylor series of the potential energy $V(x)$ of a harmonic oscillator is very special: it contains precisely one non-trivial term, that proportional to x^2 . Real oscillators have potential energy functions that invariably deviate from this ideal to some degree. The deviation is generally in the sense that $V(x) < \frac{1}{2}V''(0)x^2$ for $x > 0$ – see Figure 3.1. One reason why deviations from harmonicity are generally of this type is that it takes only a finite amount of energy to break a real object, so $V(\infty)$ should be finite, whereas the potential energy function of a harmonic oscillator increases without limit as $x \rightarrow \infty$.

Consider the anharmonic oscillator that has potential energy

$$V(x) = -\frac{a^2 V_0}{a^2 + x^2}, \quad (3.27)$$

where V_0 and a are constants. We cannot find the stationary states of this oscillator analytically any more than we can analytically solve its classical equations of motion.³ But we can determine its quantum mechanics numerically,⁴ and doing so will help to show which aspects of the results we have obtained for the harmonic oscillator are special, and which have general applicability.

³ Murphy's law is in action here: the dynamics of the pendulum is analytically intractable precisely because it is richer and more interesting than that of the harmonic oscillator.

⁴ A good way to do this is to turn the TISE into a finite matrix equation and then to use a numerical linear algebra package to find the eigenvalues of the matrix. Figure 3.2 was obtained using the approximation $\psi_n' \simeq (\psi_{n+1} + \psi_{n-1} - 2\psi_n)/\Delta^2$, where ψ_n denotes $\psi(n\Delta)$ with Δ a small increment in x . With this approximation the TISE becomes the eigenvalue equation of a tridiagonal matrix that has $2b^2/\Delta^2 + V_n/V_0$ on the leading diagonal and $-b^2/\Delta^2$ above and below this diagonal, where $b^2 = \hbar^2/2mV_0$ and $V_n = V(n\Delta)$.

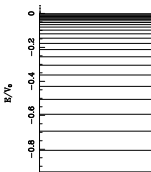


Figure 3.2 The spectrum of the anharmonic oscillator for which the potential is plotted in Figure 3.1 when the dimensionless variable $2ma^2V_0/\hbar^2 = 100$.

Figure 3.2 shows the anharmonic oscillator's energy spectrum. At low energies, when the pendulum is nearly harmonic, the energies are nearly uniformly spaced in E . As we proceed to higher energies, the spacing between levels diminishes, with the consequence that infinitely many energy levels are packed into the finite energy range between $-V_0$ and zero, where the particle becomes free. This crowding of the energy levels has the following implication for the time-dependence of $\langle x \rangle$. Suppose there are just two energies with non-zero amplitudes, a_N and a_{N+1} . Then $\langle x \rangle$ will be given by

$$\langle x \rangle = a_N^* a_{N+1} e^{i(E_N - E_{N+1})t/\hbar} \langle N | x | N+1 \rangle + \text{complex conjugate.} \quad (3.28)$$

This is a sinusoidal function of time, but its period, $T = \hbar/(E_{N+1} - E_N)$, depends on N . If we increase the energy and amplitude of the oscillator, we will increase N and Figure 3.2 shows that T will also increase. Classically the period of the oscillator increases with amplitude in just the same way. Thus there is an intimate connection between the spacing of the energy levels and classical dynamics.

Consider now the case in which the energy is more uncertain, so that several of the a_j are non-zero, and let these non-zero a_j be clustered around $j = N$ (see Figure 3.3). In this case several terms will occur in the sum for $\langle x \rangle$

$$\begin{aligned} \langle x \rangle = & \dots + a_{N-1}^* a_N e^{i(E_{N-1} - E_N)t/\hbar} \langle N-1 | x | N \rangle \\ & + a_{N+1}^* a_N e^{i(E_{N+1} - E_N)t/\hbar} \langle N+1 | x | N \rangle \\ & + a_{N+3}^* a_N e^{i(E_{N+3} - E_N)t/\hbar} \langle N+3 | x | N \rangle + \dots \end{aligned} \quad (3.29)$$

where we have anticipated a result of §4.1.4 below that the matrix element $\langle j | x | k \rangle$ vanishes if $j - k$ is even. The sum (3.29) differs from the corresponding one (3.26a) for a harmonic oscillator in the presence of matrix

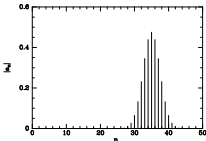


Figure 3.3 Values of a_j when there is significant uncertainty in E .

elements $\langle j|x|k\rangle$ with $|j-k| > 1$: in the case of the harmonic oscillator these matrix elements vanish, but in the general case they won't. In consequence the series contains terms with frequencies $(E_{N+3} - E_N)/\hbar$ as well as terms in $\omega_N \equiv (E_{N+1} - E_N)/\hbar$. If these additional frequencies were all integer multiples of a single frequency ω_N , the time dependence of $\langle x\rangle$ would be periodic with period $T_N = 2\pi/\omega_N$, but anharmonic, like that of the classical oscillator. Now $(E_{N+3} - E_N)/\hbar \simeq 3\omega_N$ because the spacing between energy levels changes only slowly with N , so when, as in Figure 3.3, the non-zero amplitudes are very tightly clustered around N , the additional frequencies will be integer multiples of ω_N to good accuracy, and the motion will indeed be periodic but anharmonic as classical mechanics predicts.

If we release the oscillator from near some large extension X , the non-negligible amplitudes a_j will be clustered around some integer N as depicted in Figure 3.3, and their phases will be such that at $t = 0$ the wavefunctions $\langle x|j\rangle$ will interfere constructively near X and sum to near zero elsewhere, ensuring that the mod-square of the wavefunction $\psi(x, 0) = \sum_j a_j \langle x|j\rangle$ is sharply peaked around $x = X$. At a general time the wavefunction will be given by

$$\psi(x, t) = e^{-iE_N t/\hbar} \sum_j e^{i(E_N - E_j)t/\hbar} a_j \langle x|j\rangle. \quad (3.30)$$

Since the spacing of the energy levels varies with index j , the frequencies in this sum will not be precisely equal to integer multiples of $\omega_N = (E_{N+1} - E_N)/\hbar$, so after an approximate period $T_N = 2\pi/\omega_N$ most terms in the series will not have quite returned to their values at $t = 0$. Consequently, the constructive interference around $x = X$ will be less sharply peaked than it was at $t = 0$, and the cancellation elsewhere will be correspondingly less complete. After each further approximate period T_N , the failure of terms in the series to return to their original values will be more marked, and the peak in $|\psi(x, t)|^2$ will be wider. After a long time $t \gg T_N$ the

instants at which individual terms next return to their original values will be pretty uniformly distributed around an interval in t of length T_N , and $|\psi(x, t)|^2$ will cease to evolve very much: it will have become a smooth function throughout the range $|x| \lesssim X$.

This behaviour makes perfectly good sense classically. The uncertainty in E that enables the wavefunction to be highly localised at $t = 0$ corresponds in the classical picture to uncertainty in the initial displacement X . Since the period of an anharmonic oscillator is a function of the oscillator's energy, uncertainty in X implies uncertainty in the oscillator's period. After a long time even a small uncertainty in the period translates into a significant uncertainty in the oscillator's phase. Hence after a long time the probability distribution for the particle's position is fairly uniformly distributed within $|x| \lesssim X$ even in the classical case.

Problems

3.1 After choosing units in which everything, including $\hbar = 1$, the Hamiltonian of a harmonic oscillator may be written $H = \frac{1}{2}(p^2 + x^2)$, where $[x, p] = i$. Show that if $|\psi\rangle$ is a ket that satisfies $H|\psi\rangle = E|\psi\rangle$, then

$$\frac{1}{2}(p^2 + x^2)(x \mp ip)|\psi\rangle = (E \pm 1)(x \mp ip)|\psi\rangle. \quad (3.31)$$

Explain how this algebra enables one to determine the energy eigenvalues of a harmonic oscillator.

3.2 Given that $A|E_n\rangle = \alpha|E_{n-1}\rangle$ and $E_n = (n + \frac{1}{2})\hbar\omega$, where the annihilation operator of the harmonic oscillator is

$$A = \frac{m\omega x + ip}{\sqrt{2m\hbar\omega}}, \quad (3.32)$$

show that $\alpha = \sqrt{n}$. Hint: consider $|A|E_n\rangle|^2$.

3.3 The pendulum of a grandfather clock has a period of 1 s and makes excursions of 3 cm either side of dead centre. Given that the bob weighs 0.2 kg, around what value of n would you expect its non-negligible quantum amplitudes to cluster?

3.4 Show that the minimum value of $E(p, x) = p^2/2m + \frac{1}{2}m\omega^2 x^2$ with respect to the real numbers p, x when they are constrained to satisfy $xp = \frac{1}{2}\hbar$, is $\frac{1}{2}\hbar\omega$. Explain the physical significance of this result.

3.5 How many nodes are there in the wavefunction $\langle x|n\rangle$ of the n^{th} excited state of a harmonic oscillator?

3.6 Show that in terms of a harmonic oscillator's characteristic length $\ell = \sqrt{\hbar/2m\omega}$ the ladder operators can be written

$$A = \frac{x}{2\ell} + \ell \frac{\partial}{\partial x} \quad \text{and} \quad A^\dagger = \frac{x}{2\ell} - \ell \frac{\partial}{\partial x}. \quad (3.33)$$

Hence show that the wavefunction of the second excited state is $\langle x|2\rangle = \text{constant} \times (x^2/\ell^2 - 1)e^{-x^2/4\ell^2}$ and find the normalising constant.

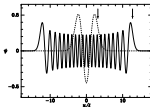


Figure 3.4 The wavefunctions $\langle x|2\rangle$ and $\langle x|40\rangle$ of two stationary states of a harmonic oscillator.

3.7 Explain why the wavefunction $\langle x|n\rangle$ of the oscillator's n^{th} stationary state must have the form

$$\langle x|n\rangle = H_n(x/\ell) e^{-x^2/4\ell^2}, \quad (3.34)$$

where H_n is an n^{th} -order ('Hermite') polynomial. By casting the equations $A|n\rangle = \sqrt{n}|n-1\rangle$ and $A^\dagger|n-1\rangle = \sqrt{n}|n\rangle$ in the x -representation, show that

$$H'_n(x/\ell) = \sqrt{n}H_{n-1}(x/\ell) \quad \text{and} \quad \sqrt{n}H_n(x/\ell) = \frac{x}{\ell}H_{n-1}(x/\ell) - H'_{n-1}(x/\ell). \quad (3.35)$$

and thus that

$$\sqrt{n}H_n(x/\ell) = \frac{x}{\ell}H_{n-1}(x/\ell) - \sqrt{n-1}H_{n-2}(x/\ell). \quad (3.36)$$

Given that $H_0 = (2\pi\ell^2)^{-1/4}$ and $H_1(y) = y/(2\pi\ell^2)^{1/4}$, use this recurrence relation to reproduce the plots of the wavefunctions $\langle x|2\rangle$ and $\langle x|40\rangle$ shown in Figure 3.4. Explain the physical significance of the vertical arrows. Why is the amplitude of $\langle x|40\rangle$ largest near the right arrow?

3.8 Use

$$x = \sqrt{\frac{\hbar}{2m\omega}}(A + A^\dagger) = \ell(A + A^\dagger) \quad (3.37)$$

to show for a harmonic oscillator that in the energy representation the operator x is

$$x_{jk} = \ell \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ \sqrt{1} & 0 & \sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & \cdots \\ & & \sqrt{3} & \cdots & \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ & & 0 & \sqrt{n-1} & \cdots \\ & & \sqrt{n-1} & 0 & \sqrt{n} & \cdots \\ & & & \sqrt{n} & 0 & \sqrt{n+1} & \cdots \\ & & & & \sqrt{n+1} & 0 & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{pmatrix} \quad (3.38)$$

Calculate the same entries for the matrix p_{jk} .

3.9 Show that the momentum operator of a harmonic oscillator can be expressed in terms of the creation and annihilation operators as

$$p = \frac{i\hbar}{2\ell}(A^\dagger - A) \quad \text{where} \quad \ell = \sqrt{\frac{\hbar}{2m\omega}}. \quad (3.39)$$

Hence show that

$$\langle 0|p^2|0\rangle = \left(\frac{\hbar}{2\ell}\right)^2. \quad (3.40)$$

How does this result relate to the physics of a free particle discussed in §2.3.3?

3.10 At $t = 0$ the state of a harmonic oscillator, mass m , frequency ω , is

$$|\psi\rangle = \frac{1}{\sqrt{2}}|N-1\rangle + \frac{1}{\sqrt{2}}|N\rangle. \quad (3.41)$$

Show that subsequently

$$\langle x \rangle_t = \sqrt{N}\ell \cos(\omega t) \quad \text{where} \quad \ell = \sqrt{\frac{\hbar}{2m\omega}}. \quad (3.42)$$

Interpret this result physically. What does this example teach us about the validity of classical mechanics?

Show that a classical oscillator with energy $(N + \frac{1}{2})\hbar\omega$ has amplitude

$$x_{\max} = 2\sqrt{N + \frac{1}{2}}\ell. \quad (3.43)$$

To explain the discrepancy between these results, consider the case in which initially

$$|\psi\rangle = \frac{1}{\sqrt{K}} \sum_{k=N}^{N+K-1} |k\rangle \quad (3.44)$$

with $N \gg K \gg 1$. Show that then $\langle x \rangle_t \simeq 2\sqrt{N}\ell \cos(\omega t)$ consistent with classical physics.

3.11* By expressing the annihilation operator A of the harmonic oscillator in the momentum representation, obtain $\langle p|0\rangle$. Check that your expression agrees with that obtained from the Fourier transform of

$$\langle x|0\rangle = \frac{1}{(2\pi\ell^2)^{1/4}} e^{-x^2/4\ell^2}, \quad \text{where} \quad \ell = \sqrt{\frac{\hbar}{2m\omega}}. \quad (3.45)$$

3.12 Show that for any two $N \times N$ matrices A, B , $\text{trace}([A, B]) = 0$. Comment on this result in the light of the results of Problem 3.8 and the canonical commutation relation $[x, p] = i\hbar$.

3.13* A **Fermi oscillator** has Hamiltonian $H = f^\dagger f$, where f is an operator that satisfies

$$f^2 = 0, \quad f f^\dagger + f^\dagger f = 1. \quad (3.46)$$

Show that $H^2 = H$, and thus find the eigenvalues of H . If the ket $|0\rangle$ satisfies $H|0\rangle = 0$ with $\langle 0|0\rangle = 1$, what are the kets (a) $|a\rangle \equiv f|0\rangle$, and (b) $|b\rangle \equiv f^\dagger|0\rangle$?

In quantum field theory the vacuum is pictured as an assembly of oscillators, one for each possible value of the momentum of each particle type. A boson is an excitation of a harmonic oscillator, while a fermion is an excitation of a Fermi oscillator. Explain the connection between the spectrum of $f^\dagger f$ and the Pauli principle.

3.14 In the time interval $(t + \delta t, t)$ the Hamiltonian H of some system varies in such a way that $|H|\psi\rangle|$ remains finite. Show that under these circumstances $|\psi\rangle$ is a continuous function of time.

A harmonic oscillator with frequency ω is in its ground state when the stiffness of the spring is instantaneously reduced by a factor $f^4 < 1$, so its natural frequency becomes $f^2\omega$. What is the probability that the oscillator is subsequently found to have energy $\frac{3}{2}\hbar f^2\omega$? Discuss the classical analogue of this problem.

3.15* P is the probability that at the end of the experiment described in Problem 3.14, the oscillator is in its second excited state. Show that when $f = \frac{1}{2}$, $P = 0.144$ as follows. First show that the annihilation operator of the original oscillator

$$A = \frac{1}{2} \{ (f^{-1} + f)A' + (f^{-1} - f)A'^\dagger \}, \quad (3.47)$$

where A' and $A'^\dagger$ are the annihilation and creation operators of the final oscillator. Then writing the ground-state ket of the original oscillator as a sum $|0\rangle = \sum_n c_n |n'\rangle$ over the energy eigenkets of the final oscillator, show that the condition $A|0\rangle = 0$ yields the recurrence relation

$$c_{n+1} = -\frac{f^{-1} - f}{f^{-1} + f} \sqrt{\frac{n}{n+1}} c_{n-1}. \quad (3.48)$$

Finally using the normalisation of $|0\rangle$, show numerically that $c_2 \simeq 0.3795$. What value do you get for the probability of the oscillator remaining in the ground state?

Show that at the end of the experiment the expectation value of the energy is $0.2656\hbar\omega$. Explain physically why this is less than the original ground-state energy $\frac{1}{2}\hbar\omega$.

This example contains the physics behind the inflationary origin of the universe: gravity explosively enlarges the vacuum, which is an infinite collection of harmonic oscillators (Problem 3.13). Excitations of these oscillators correspond to elementary particles. Before inflation the vacuum is unexcited so every oscillator is in its ground state. At the end of inflation, there is non-negligible probability of many oscillators being excited and each excitation implies the existence of a newly created particle.

3.16* In terms of the usual ladder operators A , $A^\dagger$, a Hamiltonian can be written

$$H = \mu A^\dagger A + \lambda(A + A^\dagger). \quad (3.49)$$

What restrictions on the values of the numbers μ and λ follow from the requirement for H to be Hermitian?

Show that for a suitably chosen operator B , H can be rewritten

$$H = \mu B^\dagger B + \text{constant}, \quad (3.50)$$

where $[B, B^\dagger] = 1$. Hence determine the spectrum of H .

3.17* Numerically calculate the spectrum of the anharmonic oscillator shown in Figure 3.2. From it estimate the period at a sequence of energies. Compare your quantum results with the equivalent classical results.

3.18* Let $B = cA + sA^\dagger$, where $c = \cosh \theta$, $s = \sinh \theta$ with θ a real constant and A , $A^\dagger$ are the usual ladder operators. Show that $[B, B^\dagger] = 1$.

Consider the Hamiltonian

$$H = \epsilon A^\dagger A + \frac{1}{2}\lambda(A^\dagger A^\dagger + AA), \quad (3.51)$$

where ϵ and λ are real and such that $\epsilon > \lambda > 0$. Show that when

$$\epsilon c - \lambda s = E c, \quad \lambda c - \epsilon s = E s \quad (3.52)$$

with E a constant, $[B, H] = EB$. Hence determine the spectrum of H in terms of ϵ and λ .

4

Transformations and observables

In §2.1 we associated an operator with every observable quantity through a sum over all states in which the system has a well-defined value of the observable (eq. 2.5). We found that this operator enabled us to calculate the expectation value of any function of the observable. Moreover, from the operator we could recover the observable's allowed values and the associated states because they are the operator's eigenvalues and eigenkets. These properties make an observable's operator a useful repository of information about the observable, a handy filing system. But they do not give the operator much physical meaning. Above all, they don't answer the question "what does an operator actually *do* when it operates?" In this chapter we answer this question. In the process of doing this, we will see why the canonical commutation relations (2.54) have the form that they do, and introduce the angular-momentum operators, which will play important roles in the rest of the book.

4.1 Transforming kets

When one meets an unfamiliar object, one may study it by moving it around, perhaps turning it over in one's hands so as to learn about its shape. In §1.4 we claimed that all physical information about any system is encapsulated in its ket $|\psi\rangle$, so we must learn how $|\psi\rangle$ changes as we move and turn the system.

Even the simplest systems can have orientations in addition to positions. For example, an electron, a lithium nucleus or a water molecule all

have orientations because they are not spherically symmetric: an electron is a magnetic dipole, a ${}^7\text{Li}$ nucleus has an electric quadrupole, and a water molecule is a V-shaped thing. The ket $|\psi\rangle$ that describes any of these objects contains information about the object's orientation in addition to its position and momentum. In the next subsection we shall focus on the location of a quantum system, but later we shall be concerned with its orientation as well, and in preparation for that work we explicitly display a label μ of the system's orientation and any other relevant properties, such as internal energy. For the moment μ is just an abstract symbol for orientation information; the details will be fleshed out in §7.1.

4.1.1 Translating kets

We now focus on the location of our system. To keep track of this we use a coordinate system Σ_0 whose origin is some well-defined point, say the centre of our laboratory. We can investigate $|\psi\rangle$ by expanding it in terms of a complete set of eigenstates $|\mathbf{x}, \mu\rangle$, where $\mathbf{x}$ is the position vector of the centre of mass and μ represents the system's orientation. The amplitude for finding the system's centre of mass at $\mathbf{x}$ with the orientation specified by μ is

$$\psi_\mu(\mathbf{x}) = \langle \mathbf{x}, \mu | \psi \rangle. \quad (4.1)$$

If we know all the derivatives of the wavefunction ψ_μ at a position $\mathbf{x}$, Taylor's theorem gives the value of the wavefunction at some other location $\mathbf{x} - \mathbf{a}$ as

$$\begin{aligned} \psi_\mu(\mathbf{x} - \mathbf{a}) &= \left[1 - \mathbf{a} \cdot \frac{\partial}{\partial \mathbf{x}} + \frac{1}{2!} \left(\mathbf{a} \cdot \frac{\partial}{\partial \mathbf{x}} \right)^2 - \dots \right] \psi_\mu(\mathbf{x}) \\ &= \exp \left(-\mathbf{a} \cdot \frac{\partial}{\partial \mathbf{x}} \right) \psi_\mu(\mathbf{x}) \\ &= \langle \mathbf{x}, \mu | \exp \left(-i \frac{\mathbf{a} \cdot \mathbf{p}}{\hbar} \right) | \psi \rangle. \end{aligned} \quad (4.2)$$

This equation tells us that in the state $|\psi\rangle$, the amplitude to find the system at $\mathbf{x} - \mathbf{a}$ with orientation, etc., μ is the same as the amplitude to find the system with unchanged orientation at $\mathbf{x}$ when it is in a *different* state, namely

$$|\psi'\rangle = U(\mathbf{a})|\psi\rangle \quad \text{where} \quad U(\mathbf{a}) = \exp(-i\mathbf{a} \cdot \mathbf{p}/\hbar). \quad (4.3)$$

In this notation, equation (4.2) becomes

$$\psi_\mu(\mathbf{x} - \mathbf{a}) = \langle \mathbf{x}, \mu | U(\mathbf{a}) | \psi \rangle = \langle \mathbf{x}, \mu | \psi' \rangle = \psi'_\mu(\mathbf{x}), \quad (4.4)$$

so, as Figure 4.1 illustrates, the wavefunction ψ'_μ for $|\psi'\rangle$ is the wavefunction we would expect for a system that is identical with the one described by $|\psi\rangle$ except for being shifted along the vector $\mathbf{a}$. We shall refer to this new system as the **translated** or transformed system and we shall say that

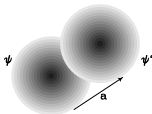


Figure 4.1 A spherical wavefunction and its displaced version.

the **translation operator** $U(\mathbf{a})$ translates $|\psi\rangle$ through $\mathbf{a}$ even though $|\psi\rangle$ is not an object in real space, so this is a slight abuse of language.¹

The ket $|\psi'\rangle$ of the translated system is a function of the vector $\mathbf{a}$. It is instructive to take its partial derivative with respect to one component of $\mathbf{a}$, say a_x . Evaluating the resulting derivative at $\mathbf{a} = 0$, when $|\psi'\rangle = |\psi\rangle$, we find

$$i\hbar \frac{\partial |\psi'\rangle}{\partial a_x} = -i\hbar \frac{\partial |\psi\rangle}{\partial x} = p_x |\psi\rangle. \quad (4.5)$$

Thus the operator p_x gives the rate at which the system's ket changes as we translate the system along the x -axis. So we have answered the question posed above as to what an observable's operator actually does in the case of the momentum operators.

Equation (2.79) enables us to expand a state of well-defined position $\mathbf{x}_0$ in terms of momentum eigenstates. We have

$$|\mathbf{x}_0, \mu\rangle = \int d^3\mathbf{p} |\mathbf{p}, \mu\rangle \langle \mathbf{p}, \mu | \mathbf{x}_0, \mu\rangle = \frac{1}{h^{3/2}} \int d^3\mathbf{p} e^{-i\mathbf{x}_0 \cdot \mathbf{p}/\hbar} |\mathbf{p}, \mu\rangle. \quad (4.6)$$

Applying the translation operator, we obtain with (4.3)

$$\begin{aligned} U(\mathbf{a})|\mathbf{x}_0, \mu\rangle &= \frac{1}{h^{3/2}} \int d^3\mathbf{p} e^{-i\mathbf{x}_0 \cdot \mathbf{p}/\hbar} U(\mathbf{a})|\mathbf{p}, \mu\rangle \\ &= \frac{1}{h^{3/2}} \int d^3\mathbf{p} e^{-i(\mathbf{x}_0 + \mathbf{a}) \cdot \mathbf{p}/\hbar} |\mathbf{p}, \mu\rangle \\ &= |\mathbf{x}_0 + \mathbf{a}, \mu\rangle, \end{aligned} \quad (4.7)$$

which is a new state, in which the system is definitely located at $\mathbf{x}_0 + \mathbf{a}$, as we would expect.

¹ An alternative point of view is that $U(\mathbf{a})|\psi\rangle$ is the same state as $|\psi\rangle$ but referred to a different coordinate system: one in which the origin lies at the point denote $\mathbf{x} = -\mathbf{a}$ in the original coordinate system. See Box 4.1 for a discussion of the relation between the 'active' transformation we consider here and coordinate transformations.

Box 4.1: Passive transformations

We can describe objects such as atoms equally well using any coordinate system. Imagine a whole family of coordinate systems set up throughout space, such that every physical point is at the origin of one coordinate system. We label by $\Sigma_{\mathbf{y}}$ the coordinate system whose origin coincides with the point labelled by $\mathbf{y}$ in our original coordinate system Σ_0 , and we indicate the coordinate system used to obtain a wavefunction by making $\mathbf{y}$ a second argument of the wavefunction; $\psi_{\mu}(\mathbf{x}; \mathbf{y})$ is the amplitude to find the system at the point labelled $\mathbf{x}$ in $\Sigma_{\mathbf{y}}$. Because the different coordinate systems vary smoothly with $\mathbf{y}$, we can use Taylor's theorem to express amplitudes in, say, $\Sigma_{\mathbf{a}+\mathbf{y}}$ in terms of amplitudes in $\Sigma_{\mathbf{y}}$. We have

$$\psi_{\mu}(\mathbf{x}; \mathbf{a} + \mathbf{y}) = \exp\left(\mathbf{a} \cdot \frac{\partial}{\partial \mathbf{y}}\right) \psi_{\mu}(\mathbf{x}; \mathbf{y}). \quad (1)$$

Now $\psi_{\mu}(\mathbf{x}; \mathbf{a}) = \psi_{\mu}(\mathbf{x} + \mathbf{a}; 0)$ because both expressions give the amplitude for the system to be at the same physical location, the point called $\mathbf{x}$ in $\Sigma_{\mathbf{a}}$ and called $\mathbf{x} + \mathbf{a}$ in Σ_0 . Then equations (4.4) and (1) give

$$(\mathbf{x}, \mu; 0|U(-\mathbf{a})|\psi) = \psi_{\mu}(\mathbf{x} + \mathbf{a}; 0) = \psi_{\mu}(\mathbf{x}; \mathbf{a}), \quad (2)$$

where again $|\mathbf{x}, \mu; 0\rangle$ indicates the state in which the system is located at the point labelled by $\mathbf{x}$ in Σ_0 . This equation tells us that $|\psi\rangle$ has the same wavefunction in $\Sigma_{\mathbf{a}}$ that $U(-\mathbf{a})|\psi\rangle$ has in Σ_0 . Therefore, moving the origin of our coordinates through a vector $\mathbf{a}$ has the same effect on an arbitrary state's wavefunction as moving the system itself through $-\mathbf{a}$. Physically moving the system is known as an **active transformation**, whereas leaving the state alone but changing the coordinate system is called a **passive transformation**. The infinitesimal vectors required to make logically equivalent active and passive transformations differ in sign. This sign difference reflects the fact that if you move backwards, the world around you seems to move forwards; hence moving the origin of one's coordinates back by $\delta\mathbf{a}$ has the same effect as moving the system forward by $\delta\mathbf{a}$. In this book we confine ourselves to active transformations.

4.1.2 Continuous transformations and generators

In §1.4 we saw that the normalisation condition $\langle\psi|\psi\rangle = 1$ expresses the fact that when we make a measurement of any observable, we will measure *some* value; for example, if we determine the system's location, we will find it *somewhere*. The normalisation condition must be unaffected by any transformation that we make on a system, so the transformation operator² U must have the property that for any state $|\psi\rangle$

$$1 = \langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle. \quad (4.8)$$

² We restrict ourselves to the case in which the operator U is linear, as is every operator used in this book. In consequence, we are unable to consider time reversal.

Box 4.2: Operators from expectation values

In this box we show that if

$$\langle \psi | A | \psi \rangle = \langle \psi | B | \psi \rangle, \quad (1)$$

for every state $|\psi\rangle$, then the operators A and B are identical. We set $|\psi\rangle = |\phi\rangle + \lambda|\chi\rangle$, where λ is a complex number. Then equation (1) implies

$$\lambda(\langle \phi | A | \chi \rangle - \langle \phi | B | \chi \rangle) = \lambda^* (\langle \chi | B | \phi \rangle - \langle \chi | A | \phi \rangle). \quad (2)$$

Since equation (1) is valid for any state $|\psi\rangle$, equation (2) remains valid as we vary λ . If the coefficients of λ and λ^* are non-zero, we can cause the left and right sides of (2) to change differently by varying the phase of λ ; they can be equal irrespective of the phase of λ only if the coefficients vanish. This shows that $\langle \chi | A | \phi \rangle = \langle \chi | B | \phi \rangle$ for arbitrary states $|\phi\rangle$ and $|\chi\rangle$, from which it follows that $A = B$.

From this requirement we can infer by the argument given in Box 4.2 (with $A = U^\dagger U$ and $B = I$, the identity operator) that $U^\dagger U = I$, so $U^\dagger = U^{-1}$. Operators with this property are called **unitary operators**. When we transform all states with a unitary operator, we leave unchanged all amplitudes: $\langle \phi' | \psi' \rangle = \langle \phi | \psi \rangle$ for any states $|\phi\rangle$ and $|\psi\rangle$.

Exactly how we construct a unitary operator depends on the type of transformation we wish it to make. The identity operator is the unitary operator that represents doing nothing to our system. The translation operator $U(\mathbf{a})$ can be made to approach the identity as closely as we please by diminishing the magnitude of $\mathbf{a}$. Many other unitary operators also have a parameter θ that can be reduced to zero such that the operator tends to the identity. In this case we can write for small $\delta\theta$

$$U(\delta\theta) = I - i\delta\theta \tau + O(\delta\theta)^2, \quad (4.9)$$

where the factor of i is a matter of convention and τ is an operator. The unitarity of U implies that

$$I = U^\dagger(\delta\theta)U(\delta\theta) = I + i\delta\theta(\tau^\dagger - \tau) + O(\delta\theta^2). \quad (4.10)$$

Equating powers of $\delta\theta$ on the two sides of the equation, we deduce that τ is Hermitian, so it may be an observable. If so, its eigenkets are states in which the system has well-defined values of the observable τ .

We obtain an important equation by using equation (4.9) to evaluate $|\psi'\rangle = U(\delta\theta)|\psi\rangle$. Subtracting $|\psi\rangle$ from both sides of the resulting equation, dividing through by $\delta\theta$ and proceeding to the limit $\delta\theta \rightarrow 0$, we obtain

$$i \frac{\partial |\psi'\rangle}{\partial \theta} = \tau |\psi'\rangle. \quad (4.11)$$

Thus the observable τ gives the rate at which $|\psi\rangle$ changes when we increase the parameter θ in the unitary transformation that τ generates. Equation (4.5) is a concrete example of this equation in action.

A finite transformation can be generated by repeatedly performing an infinitesimal one. Specifically, if we transform N times with $U(\delta\theta)$ with $\delta\theta = \theta/N$, then in the limit $N \rightarrow \infty$ we have

$$U(\theta) = \lim_{N \rightarrow \infty} \left(1 - i \frac{\theta}{N} \tau \right)^N = e^{-i\theta\tau}. \quad (4.12)$$

This relation is clearly a generalisation of the definition (4.3) of the translation operator. The Hermitian operator τ is called the **generator** of both the unitary operator U and the transformations that U accomplishes; for example, $\mathbf{p}/\hbar$ is the generator of translations.

4.1.3 The rotation operator

Consider what happens if we rotate the system. Whereas in §4.1.1 we constructed a state $|\psi'\rangle = U(\mathbf{a})|\psi\rangle$ that differed from the state $|\psi\rangle$ only in a shift by $\mathbf{a}$ in the location of the centre of mass, we now wish to find a rotation operator that constructs the state $|\psi'\rangle$ that we would get if we could somehow rotate the apparatus on a turntable without disturbing its internal structure in any way. Whereas the orientation of the system is unaffected by a translation, it will be changed by the rotation operator, as is physically evident if we imagine turning a non-spherical object on a turntable.

From §4.1.2 we know that a rotation operator will be unitary, and have a Hermitian generator. Actually, we expect there to be several generators, just as there are three generators, $p_x/\hbar$, $p_y/\hbar$ and $p_z/\hbar$, of translations. Because there are three generators of translations, three numbers, the components of the vector $\mathbf{a}$ in equation (4.3), are required to specify a particular translation. Hence we anticipate that the number of generators of rotations will equal the number of angles that are required to specify a rotation. Two angles are required to specify the axis of rotation, and a third is required to specify the angle through which we rotate. Thus by analogy with equation (4.3), we expect that a general rotation operator can be obtained by exponentiating a linear combination of three generators of rotations, and we write

$$U(\boldsymbol{\alpha}) = \exp(-i\boldsymbol{\alpha} \cdot \mathbf{J}). \quad (4.13)$$

Here $\boldsymbol{\alpha}$ is a vector that specifies a rotation through an angle $|\boldsymbol{\alpha}|$ around the direction of the unit vector $\hat{\boldsymbol{\alpha}}$, and $\mathbf{J}$ is comprised of three Hermitian operators, J_x , J_y and J_z . In the course of this chapter and the next it will become clear that the observable associated with $\mathbf{J}$ is angular momentum. Consequently, the components of $\mathbf{J}$ are called the **angular-momentum operators**.

The role that the angular-momentum operators play in rotating the system around the axis $\hat{\boldsymbol{\alpha}}$ is expressed by rewriting equation (4.11) with appropriate substitutions as

$$i \frac{\partial |\psi\rangle}{\partial \alpha} = \hat{\boldsymbol{\alpha}} \cdot \mathbf{J} |\psi\rangle. \quad (4.14)$$

4.1.4 Discrete transformations

The parity operator Not all transformations are continuous. In physics, the most prominent example of a discrete transformation is the **parity transformation** $\mathcal{P}$, which swaps the sign of the coordinates of all spatial points; the action of $\mathcal{P}$ on coordinates is represented by the matrix

$$\mathcal{P} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \text{so} \quad \mathcal{P}\mathbf{x} = -\mathbf{x}. \quad (4.15)$$

Notice that $\det \mathcal{P} = -1$, whereas a rotation matrix has $\det \mathbf{R} = +1$. In fact, any linear transformation with determinant equal to -1 can be written as a product of $\mathcal{P}$ and a rotation $\mathbf{R}$.

Let some quantum state $|\psi\rangle$ have wavefunction $\psi_\mu(\mathbf{x}) = \langle \mathbf{x}, \mu | \psi \rangle$, where the label μ is the usual shorthand for the system's orientation. Then the quantum parity operator P is defined by

$$\psi'_\mu(\mathbf{x}) = \langle \mathbf{x}, \mu | P | \psi \rangle = \psi_\mu(\mathcal{P}\mathbf{x}) = \psi_\mu(-\mathbf{x}) = \langle -\mathbf{x}, \mu | \psi \rangle. \quad (4.16)$$

The wavefunction of the new state, $|\psi'\rangle = P|\psi\rangle$, takes the same value at $\mathbf{x}$ that the old wavefunction does at $-\mathbf{x}$. Thus, when the system is in the state $P|\psi\rangle$, it has the same amplitude to be at $\mathbf{x}$ as it had to be at $-\mathbf{x}$ when it was in the state $|\psi\rangle$. The orientation and internal properties of the system are unaffected by P . The invariance of orientation under a parity transformation is not self-evident, but in §4.2 we shall see that it follows from the rules that govern commutation of P with $\mathbf{x}$ and $\mathbf{J}$.

Applying the parity operator twice creates a state $|\psi''\rangle = P|\psi'\rangle = P^2|\psi\rangle$ with wavefunction

$$\begin{aligned} \psi''_\mu(\mathbf{x}) &= \langle \mathbf{x}, \mu | P | \psi' \rangle = \langle -\mathbf{x}, \mu | \psi' \rangle = \langle -\mathbf{x}, \mu | P | \psi \rangle = \langle \mathbf{x}, \mu | \psi \rangle \\ &= \psi_\mu(\mathbf{x}). \end{aligned} \quad (4.17)$$

Hence $P^2 = 1$ and an even number of applications of the parity operator leaves the wavefunction unchanged. It also follows that $P = P^{-1}$ is its own inverse.

P is also Hermitian:

$$\begin{aligned} \langle \phi | P | \psi \rangle^* &= \int d^3\mathbf{x} \sum_\mu (\langle \phi | \mathbf{x}, \mu \rangle \langle \mathbf{x}, \mu | P | \psi \rangle)^* \\ &= \int d^3\mathbf{x} \sum_\mu (\langle \phi | \mathbf{x}, \mu \rangle \langle -\mathbf{x}, \mu | \psi \rangle)^* \\ &= \int d^3\mathbf{x} \sum_\mu \langle \psi | -\mathbf{x}, \mu \rangle \langle \mathbf{x}, \mu | P | \phi \rangle \\ &= \int d^3\mathbf{x} \sum_\mu \langle \psi | -\mathbf{x}, \mu \rangle \langle -\mathbf{x}, \mu | P | \phi \rangle = \langle \psi | P | \phi \rangle, \end{aligned} \quad (4.18)$$



Figure 4.2 If the coordinates of the point marked with a filled circle are (x, y) , then the coordinates of the point marked with an open circle are (y, x) . The points would be object and image if a mirror lay along the line $y = x$.

so $P^\dagger = P$. It now follows that P is unitary³ because $P^{-1} = P = P^\dagger$. Hence from the discussion of §4.1.2 it follows that transforming all states with P will preserve all amplitudes for the system.

Suppose now that $|P\rangle$ is an eigenket of P , with eigenvalue λ . Then $|P\rangle = P^2|P\rangle = \lambda P|P\rangle = \lambda^2|P\rangle$, so $\lambda^2 = 1$. Thus the eigenvalues of P are ± 1 . Eigenstates of P are said to have **definite parity**, with $|+\rangle = P|+\rangle$ being a state of **even parity** and $|-\rangle = -P|-\rangle$ being one of **odd parity**.

In §3.1 we found that the stationary-state wavefunctions of a harmonic oscillator are even functions of x when the quantum number n is even, and odd functions of x otherwise. It is clear that these stationary states are also eigenstates of P , those for $n = 0, 2, 4, \dots$ having even parity and those for $n = 1, 3, 5, \dots$ having odd parity.

Mirror operators Systems not infrequently exhibit a mirror symmetry of some sort. When they do, it can be helpful to define an operator which transforms any state into the corresponding mirror state. Here's an illustrative concrete example.

A particle moves in two dimensions, so the amplitudes $\langle x, y|\psi\rangle$ for the particle to be found at the location (x, y) constitute a complete set of amplitudes. Let the operator M be such that for any state $|\psi\rangle$

$$\langle x, y|M|\psi\rangle = \langle y, x|\psi\rangle. \quad (4.19)$$

That is, in the state $M|\psi\rangle$ the amplitude to be at the point (x, y) (marked with a filled dot in Figure 4.2) is the same as the amplitude to be at the point (y, x) (marked by an open dot) when in the state $|\psi\rangle$. If there were a mirror along the line $y = x$, the image of a light at (x, y) would be located at (y, x) . Thus the operator M produces the state we get by mirroring all the amplitudes in the line $y = x$. We leave as an exercise (Problem 4.12) the proof that M is a unitary operator, which closely follows the proof we gave of the unitarity of P .

If we mirror a set of amplitudes twice, we obviously recover the original amplitudes, so $M^2 = 1$ and it follows that the eigenvalues of M can only be ± 1 .

³ Problem 4.9 shows that P is unitary by showing that it has an infinitesimal generator.

4.2 Transformations of operators

When we move an object around, we expect to find it in a new place. Specifically, suppose $\langle \psi | \mathbf{x} | \psi \rangle = \mathbf{x}_0$ for some state $|\psi\rangle$. Since $\mathbf{x}_0$ just labels a spatial point, it must behave under translations and rotations like any vector. For example, translating a system that is in the state $|\psi\rangle$ through $\mathbf{a}$, we obtain a new state $|\psi'\rangle$ which has $\langle \psi' | \mathbf{x} | \psi' \rangle = \mathbf{x}_0 + \mathbf{a} = \langle \psi | \mathbf{x} + I\mathbf{a} | \psi \rangle$. On the other hand, from §4.1.1 we know that $\langle \psi' | \mathbf{x} | \psi' \rangle = \langle \psi | U^\dagger(\mathbf{a}) \mathbf{x} U(\mathbf{a}) | \psi \rangle$. Since these expectation values must be equal for any initial state $|\psi\rangle$, it follows from the argument given in Box 4.2 that

$$U^\dagger(\mathbf{a}) \mathbf{x} U(\mathbf{a}) = \mathbf{x} + \mathbf{a}, \quad (4.20)$$

where the identity operator is understood to multiply the constant $\mathbf{a}$. For an infinitesimal translation with $\mathbf{a} \rightarrow \delta\mathbf{a}$ we have $U(\mathbf{a}) \simeq 1 - i\mathbf{a} \cdot \mathbf{p}/\hbar$. So

$$\begin{aligned} \mathbf{x} + \delta\mathbf{a} &\simeq \left(1 + i\frac{\delta\mathbf{a} \cdot \mathbf{p}}{\hbar}\right) \mathbf{x} \left(1 - i\frac{\delta\mathbf{a} \cdot \mathbf{p}}{\hbar}\right) \\ &= \mathbf{x} - \frac{i}{\hbar} [\mathbf{x}, \delta\mathbf{a} \cdot \mathbf{p}] + O(\delta\mathbf{a})^2. \end{aligned} \quad (4.21)$$

For this to be true for all small vectors $\delta\mathbf{a}$, $\mathbf{x}$ and $\mathbf{p}$ must satisfy the commutation relations

$$[x_i, p_j] = i\hbar\delta_{ij} \quad (4.22)$$

in accordance with equation (2.54). Here we see that this commutation relation arises as a natural consequence of the properties of $\mathbf{x}$ under translations. For a finite translation, we can write

$$U^\dagger(\mathbf{a}) \mathbf{x} U(\mathbf{a}) = U^\dagger(\mathbf{a}) U(\mathbf{a}) \mathbf{x} + U^\dagger(\mathbf{a}) [\mathbf{x}, U(\mathbf{a})] = \mathbf{x} + U^\dagger(\mathbf{a}) [\mathbf{x}, U(\mathbf{a})]. \quad (4.23)$$

We use equation (2.25) to evaluate the commutator on the right. Treating U as the function $e^{-i\mathbf{a} \cdot \mathbf{p}/\hbar}$ of $\mathbf{a} \cdot \mathbf{p}$, we find

$$U^\dagger(\mathbf{a}) \mathbf{x} U(\mathbf{a}) = \mathbf{x} - \frac{i}{\hbar} U^\dagger(\mathbf{a}) [\mathbf{x}, \mathbf{a} \cdot \mathbf{p}] U(\mathbf{a}) = \mathbf{x} + \mathbf{a} \quad (4.24)$$

as equation (4.20) requires.

Similarly, under rotations ordinary spatial vectors have components which transform as $\mathbf{v} \rightarrow \mathbf{R}(\boldsymbol{\alpha})\mathbf{v}$, where $\mathbf{R}(\boldsymbol{\alpha})$ is a matrix describing a rotation through angle $|\boldsymbol{\alpha}|$ around the $\hat{\boldsymbol{\alpha}}$ -axis. The expectation values $\langle \psi | \mathbf{x} | \psi \rangle = \mathbf{x}_0$ should then transform in this way. In §4.1.2 we saw that when a system is rotated through an angle $|\boldsymbol{\alpha}|$ around the $\hat{\boldsymbol{\alpha}}$ -axis, its ket $|\psi\rangle$ should be multiplied by $U(\boldsymbol{\alpha}) = e^{-i\boldsymbol{\alpha} \cdot \mathbf{J}}$. If this transformation of $|\psi\rangle$ is to be consistent with the rotation of the expectation value of $\mathbf{x}$, we need

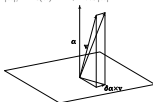
$$\mathbf{R}(\boldsymbol{\alpha}) \langle \psi | \mathbf{x} | \psi \rangle = \langle \psi' | \mathbf{x} | \psi' \rangle = \langle \psi | U^\dagger(\boldsymbol{\alpha}) \mathbf{x} U(\boldsymbol{\alpha}) | \psi \rangle. \quad (4.25)$$

Since this must hold for any state $|\psi\rangle$, from the argument given in Box 4.2 it follows that

$$\mathbf{R}(\boldsymbol{\alpha}) \mathbf{x} = U^\dagger(\boldsymbol{\alpha}) \mathbf{x} U(\boldsymbol{\alpha}). \quad (4.26)$$

Box 4.3: Rotations in ordinary space

A rotation matrix $\mathbf{R}$ is defined by the conditions $\mathbf{R}^T = \mathbf{R}^{-1}$ and $\det(\mathbf{R}) = +1$. If $\mathbf{R}(\boldsymbol{\alpha})$ rotates around the $\hat{\boldsymbol{\alpha}}$ -axis, it should leave this axis invariant so $\mathbf{R}(\boldsymbol{\alpha})\hat{\boldsymbol{\alpha}} = \hat{\boldsymbol{\alpha}}$. For a rotation through an angle $|\boldsymbol{\alpha}|$, $\text{Tr } \mathbf{R}(\boldsymbol{\alpha}) = 1 + 2 \cos |\boldsymbol{\alpha}|$.



Let $\boldsymbol{\alpha}$ be an infinitesimal rotation vector: that is, a rotation through α around the axis that is in the direction of the unit vector $\hat{\boldsymbol{\alpha}}$. We consider the effect of rotating an arbitrary vector $\mathbf{v}$ through angle α . The component of $\mathbf{v}$ parallel to $\boldsymbol{\alpha}$ is unchanged by the rotation.

The figure shows the projection of $\mathbf{v}$ into the plane perpendicular to $\boldsymbol{\alpha}$. The rotated vector is seen to be the vectorial sum of $\mathbf{v}$ and the infinitesimal vector $\boldsymbol{\alpha} \times \mathbf{v}$ that joins the end of $\mathbf{v}$ before and after rotation. That is

$$\mathbf{v}' = \mathbf{v} + \boldsymbol{\alpha} \times \mathbf{v}.$$

For an infinitesimal rotation, $\boldsymbol{\alpha} \rightarrow \delta\boldsymbol{\alpha}$ and $\mathbf{R}(\boldsymbol{\alpha})\mathbf{x} \simeq \mathbf{x} + \delta\boldsymbol{\alpha} \times \mathbf{x}$ as is shown in Box 4.3, so equation (4.26) becomes

$$\begin{aligned} \mathbf{x} + \delta\boldsymbol{\alpha} \times \mathbf{x} &\simeq (1 + i\delta\boldsymbol{\alpha} \cdot \mathbf{J}) \mathbf{x} (1 - i\delta\boldsymbol{\alpha} \cdot \mathbf{J}) \\ &= \mathbf{x} + i[\delta\boldsymbol{\alpha} \cdot \mathbf{J}, \mathbf{x}] + O(\delta\boldsymbol{\alpha})^2. \end{aligned} \quad (4.27)$$

In components, the vector product $\delta\boldsymbol{\alpha} \times \mathbf{x}$ can be written

$$(\delta\boldsymbol{\alpha} \times \mathbf{x})_i = \sum_{jk} \epsilon_{ijk} \delta\alpha_j x_k, \quad (4.28)$$

where ϵ_{ijk} is the object that changes sign if any two subscripts are interchanged and has $\epsilon_{xyz} = 1$ (Appendix B). For example, equation (4.28) gives $(\delta\boldsymbol{\alpha} \times \mathbf{x})_x = \sum_{jk} \epsilon_{xjk} \delta\alpha_j x_k = \epsilon_{xyz} \delta\alpha_y x_z + \epsilon_{xzy} \delta\alpha_z x_y = \delta\alpha_y x_z - \delta\alpha_z x_y$. The i^{th} components of equation (4.27) is

$$\sum_{jk} \epsilon_{ijk} \delta\alpha_j x_k = i \sum_j \delta\alpha_j [J_j, x_i]. \quad (4.29)$$

Since this equation holds for arbitrary $\delta\boldsymbol{\alpha}$, we conclude that the position and angular-momentum operators x_i and J_j must satisfy the commutation relation

$$[J_i, x_j] = i \sum_k \epsilon_{ijk} x_k. \quad (4.30)$$

In particular, $[J_x, y] = iz$ and $[J_z, x] = iy$, while $[J_x, x] = 0$.

In fact, if the expectation value of any operator $\mathbf{v}$ is a spatial vector, then the argument just given in the case of $\mathbf{x}$ shows that the components v_i must satisfy

$$[J_i, v_j] = i \sum_k \epsilon_{ijk} v_k. \quad (4.31)$$

For example, since momentum is a vector, equation (4.31) with $\mathbf{v} = \mathbf{p}$ gives the commutation relations of $\mathbf{p}$ with $\mathbf{J}$,

$$[J_i, p_j] = i \sum_k \epsilon_{ijk} p_k. \quad (4.32)$$

The product $\boldsymbol{\alpha} \cdot \mathbf{J}$ must be invariant under coordinate rotations because the operator $U(\boldsymbol{\alpha}) = e^{-i\boldsymbol{\alpha} \cdot \mathbf{J}}$ depends on the *direction* $\boldsymbol{\alpha}$ and not on the numbers used to quantify that direction. Since $\boldsymbol{\alpha}$ is an arbitrary vector, the invariance of $\boldsymbol{\alpha} \cdot \mathbf{J}$ under rotations implies that under rotations the components of $\mathbf{J}$ transform like those of a vector. Hence, in equation (4.31) we can replace $\mathbf{v}$ by $\mathbf{J}$ to obtain the commutation relation

$$[J_i, J_j] = i \sum_k \epsilon_{ijk} J_k. \quad (4.33)$$

In §7.1 we shall deduce the spectrum of the angular-momentum operators from this relation.

We now show that $\mathbf{J}$ commutes with any operator S whose expectation value is a scalar. The proof is simple: $\langle \psi | S | \psi \rangle$, being a scalar, is not affected by rotations, so

$$\langle \psi' | S | \psi' \rangle = \langle \psi | U^\dagger(\boldsymbol{\alpha}) S U(\boldsymbol{\alpha}) | \psi \rangle = \langle \psi | S | \psi \rangle. \quad (4.34)$$

Equating the operators on either side of the second equality and using $U^{-1} = U^\dagger$ we have $[S, U] = 0$. Restricting U to an infinitesimal rotation gives

$$S \simeq (1 + i\delta\boldsymbol{\alpha} \cdot \mathbf{J}) S (1 - i\delta\boldsymbol{\alpha} \cdot \mathbf{J}) = S + i\delta\boldsymbol{\alpha} \cdot [\mathbf{J}, S] + O(\delta\boldsymbol{\alpha})^2. \quad (4.35)$$

Since $\delta\boldsymbol{\alpha}$ is arbitrary, it follows that

$$[\mathbf{J}, S] = 0. \quad (4.36)$$

Among other things, this tells us that $[\mathbf{J}, \mathbf{x} \cdot \mathbf{x}] = [\mathbf{J}, \mathbf{p} \cdot \mathbf{p}] = [\mathbf{J}, \mathbf{x} \cdot \mathbf{p}] = 0$. It is straightforward to check that these results are consistent with the vector commutation relations (4.31) (Problem 4.1). It also follows that $J^2 = \mathbf{J} \cdot \mathbf{J}$ commutes with all of the J_i ,

$$[\mathbf{J}, J^2] = 0. \quad (4.37)$$

Equations (4.33) and (4.37) imply that it is possible to find a complete set of simultaneous eigenstates for both J^2 and any one component of $\mathbf{J}$ (but only one).

The parity operator Under a parity transform, coordinates behave as $\mathbf{x} \rightarrow \mathcal{P}\mathbf{x} = -\mathbf{x}$ whereas quantum states transform as $|\psi\rangle \rightarrow |\psi'\rangle = P|\psi\rangle$, so

$$-\langle \psi | \mathbf{x} | \psi \rangle = \mathcal{P} \langle \psi | \mathbf{x} | \psi \rangle = \langle \psi' | \mathbf{x} | \psi' \rangle = \langle \psi | P^\dagger \mathbf{x} P | \psi \rangle, \quad (4.38)$$

which implies that $P^\dagger \mathbf{x} P = -\mathbf{x}$ or, since P is a unitary operator,

$$\{\mathbf{x}, P\} = \mathbf{x}P + P\mathbf{x} = 0. \quad (4.39)$$

Two operators A and B for which $\{A, B\} = 0$ are said to **anticommute**, with $\{A, B\}$ being their **anticommutator**. The argument we have just given for $\mathbf{x}$ works with $\mathbf{x}$ replaced by any vector operator $\mathbf{v}$, so we always have

$$\{\mathbf{v}, P\} = \mathbf{v}P + P\mathbf{v} = 0. \quad (4.40)$$

This relation contains important information about the action of P . Suppose $|\omega\rangle$ is an eigenstate of a vector operator $\mathbf{v}$ with eigenvalues ω such that $\mathbf{v}|\omega\rangle = \omega|\omega\rangle$. From equation (4.40) we see that

$$\mathbf{v}|\omega'\rangle = \mathbf{v}(P|\omega\rangle) = -P\mathbf{v}|\omega\rangle = -\omega P|\omega\rangle = -\omega|\omega'\rangle \quad (4.41)$$

so the parity-reversed state $|\omega'\rangle = P|\omega\rangle$ is also an eigenstate of $\mathbf{v}$, but the eigenvalue has changed sign.

Let $|\pm\rangle$ be states of definite parity such that $P|\pm\rangle = \pm|\pm\rangle$. With equation (4.40) we deduce that

$$-(\pm|\mathbf{v}|\pm) = \mathcal{P}(\pm|\mathbf{v}|\pm) = \langle\pm|P^\dagger\mathbf{v}P|\pm\rangle = (\pm)^2\langle\pm|\mathbf{v}|\pm\rangle. \quad (4.42)$$

Since zero is the only number that is equal to minus itself, all vector operators have vanishing expectation value in states of definite parity. More generally, if $|\phi\rangle$ and $|\chi\rangle$ both have the same definite parity, equation (4.40) implies that $\langle\phi|\mathbf{v}|\chi\rangle = 0$. We'll use this result in Chapter 10.

We frequently encounter situations in which the potential energy $V(\mathbf{x})$ is an even function of $\mathbf{x}$: $V(-\mathbf{x}) = V(\mathbf{x})$. We then say that the potential is **reflection-symmetric** because the potential energy at $-\mathbf{x}$ is the same as it is at the point $\mathbf{x}$ into which $-\mathbf{x}$ is mapped by reflection through the origin. We now show that in such a case the parity operator commutes with the Hamiltonian. For an arbitrary state $|\psi\rangle$ consider the amplitude

$$\langle\mathbf{x}|PV|\psi\rangle = \langle-\mathbf{x}|V|\psi\rangle = V(-\mathbf{x})\langle-\mathbf{x}|\psi\rangle = V(\mathbf{x})\langle-\mathbf{x}|\psi\rangle, \quad (4.43a)$$

where we have used equation (4.16). On the other hand

$$\langle\mathbf{x}|VP|\psi\rangle = V(\mathbf{x})\langle\mathbf{x}|P|\psi\rangle = V(\mathbf{x})\langle-\mathbf{x}|\psi\rangle. \quad (4.43b)$$

Since $\mathbf{x}$ and $|\psi\rangle$ are arbitrary, it follows that when V is an even function of $\mathbf{x}$, $[P, V] = 0$. This argument generalises to all operators that carry out a transformation that is a symmetry of the potential energy.

Since the momentum $\mathbf{p}$ is a vector operator, $P\mathbf{p} = -\mathbf{p}P$, so

$$\begin{aligned} p^2 P &= \sum_k p_k p_k P = - \sum_k p_k P p_k = \sum_k P p_k p_k = P p^2 \\ \Rightarrow [p^2, P] &= 0. \end{aligned} \quad (4.44)$$

Applying these results to the Hamiltonian $H = p^2/2m + V(\mathbf{x})$ of a particle of mass m that moves in a reflection-symmetric potential, we have that $[H, P] = 0$. It follows that for such a particle there is a complete set of stationary states of well-defined parity. This fact is illustrated by the case of the harmonic oscillator studied in §3.1, and in Chapter 5 it will enable us dramatically to simplify our calculations.

In classical physics, a vector product $\mathbf{a} \times \mathbf{b}$ is a **pseudovector**; it behaves like an ordinary vector under rotations, but is invariant under parity, since both $\mathbf{a}$ and $\mathbf{b}$ change sign. We now show that expectation values of the angular-momentum operators, $\langle \mathbf{J} \rangle$, are pseudovectors. If v_i are components of a vector operator, then combining equations (4.31) and (4.40), we obtain

$$\{P, [v_i, J_j]\} = i \sum_k \epsilon_{ijk} \{P, v_k\} = 0. \quad (4.45)$$

We use the identity (4.77) proved in Problem 4.8 to rewrite the left side of this equation. We obtain

$$0 = \{P, [v_i, J_j]\} = \{[P, v_i], J_j\} - \{[P, J_j], v_i\} = -\{[P, J_j], v_i\}. \quad (4.46)$$

Hence the operator $[P, J_j]$ anticommutes with any component of an arbitrary vector. Since P is defined to have precisely this property, $[P, J_j]$ must be proportional to P , that is

$$[P, J_j] = \lambda P, \quad (4.47)$$

where λ must be the same for all values of j because the three coordinate directions are equivalent. Under rotations, the left side transforms like a vector, while the right side is invariant. This is possible only if both sides vanish. Hence the parity operator commutes with all three angular-momentum operators. It now follows that

$$\langle \psi' | \mathbf{J} | \psi' \rangle = \langle \psi | P^\dagger \mathbf{J} P | \psi \rangle = \langle \psi | \mathbf{J} | \psi \rangle, \quad (4.48)$$

so $\langle \mathbf{J} \rangle$ is unchanged by a parity transformation, and is a pseudovector.

Mirror operators In §4.1.4 we introduced a typical mirror operator M . To discover how M interacts with the position operators x and y we argue that for any state $|\psi\rangle$

$$\langle \psi | M^\dagger x M | \psi \rangle = \langle \psi | y | \psi \rangle. \quad (4.49)$$

That is, in the state $M|\psi\rangle$ the expectation of x must be equal to the expectation value of y in the state $|\psi\rangle$ – the truth of this statement follows immediately from the definition (4.19) of the state $M|\psi\rangle$. Since equation (4.49) holds for arbitrary $|\psi\rangle$, we can infer the operator equation

$$M^\dagger x M = y \quad \Rightarrow \quad x M = M y, \quad (4.50)$$

where the second equation follows by multiplying both sides of the first equation by M and using the unitarity condition $M M^\dagger = I$. In the same way we can show that $M x = y M$, $p_x M = M p_y$ and $p_y M = M p_x$.

4.3 Symmetries and conservation laws

Time changes states: in a given time interval t , the natural evolution of the system causes any state $|\psi, 0\rangle$ to evolve to another state $|\psi, t\rangle$. Equation (2.32) gives an explicit expression for $|\psi, t\rangle$. It is easy to see that with the present notation this rule can be written

$$|\psi, t\rangle = e^{-iHt/\hbar} |\psi, 0\rangle, \quad (4.51)$$

where H is the Hamiltonian. The **time-evolution operator**

$$U(t) = e^{-iHt/\hbar} \quad (4.52)$$

is unitary, as we would expect.⁴

Now suppose that the generator τ of some displacement (a translation, a rotation, or something similar) commutes with H . Since these operators commute, their exponentials $U(\theta)$ (eq. 4.12) and $U(t)$ also commute. Consequently, for any state $|\psi\rangle$

$$U(\theta)U(t)|\psi\rangle = U(t)U(\theta)|\psi\rangle. \quad (4.53)$$

The left side is the state you get by waiting for time t and then displacing, while the right side is the state obtained by displacing first and then waiting. So the equation says that the system evolves in the same way no matter where you put it. That is, there is a connection between observables that commute with the Hamiltonian and invariance of the physics under displacements. Moreover, if at $t = 0$ the system is in an eigenstate $|\tau_n\rangle$ of the operator τ , then at a later time t it is still in an eigenstate of τ with the same eigenvalue because

$$\tau(U(t)|\tau_n\rangle) = U(t)\tau|\tau_n\rangle = \tau_n(U(t)|\tau_n\rangle). \quad (4.54)$$

This result confirms the conclusion we reached in §2.2.1 that observables that commute with H provide ‘good quantum numbers’ and assures us that whenever the physics is unchanged by a displacement, there is a conserved quantity.

If $[p_x, H] = 0$, this argument implies that the system evolves in the same way wherever it is located. We say that the Hamiltonian is **translationally** invariant. It is a fundamental premise of physics that empty space is the same everywhere, so the Hamiltonian of every isolated system is translationally invariant. Consequently, when a system is isolated, the expectation value of any function of the momentum operators is a

⁴The similarity between equations (4.52) and the formula (4.12) for a general unitary transformation suggests that H is the generator of transformations in time. This is not quite true. If we were to push the system forward in time in the same way that we translate it in x , we would delay the instant at which we would impose some given initial conditions, with the result that it would be *less* evolved at a given time t . The time-evolution operator, by contrast, makes the system older. Hence H is the generator of transformations backwards in time.

conserved quantity, and, if the system is started in a state of well-defined momentum, it will stay in that state. This is Newton's first law.

If $[J_z, H] = 0$, we say that the Hamiltonian is rotationally invariant around the z -axis, and our argument implies that the system evolves in the same way no matter how it is turned around the z -axis. The expectation value of any function of J_z is constant, and if the state is initially in an eigenstate of J_z with eigenvalue m , it will remain in that state. Consequently, m is a good quantum number. In classical physics invariance of a system's dynamics under rotations around the z -axis is associated with conservation of the z component of the system's angular momentum. This fact inspires the identification of $\hbar \mathbf{J}$ with angular momentum.

Above we used a very general argument to infer that the existence of a unitary operator that commutes with the Hamiltonian implies that the system has a symmetry. In §4.1.4 an explicit calculation (eq. 4.43) showed that reflection symmetry of the potential energy implied that the potential energy operator V commutes with the parity operator P . This argument generalises to other transformation operators. For example, suppose $V(\mathbf{x})$ is invariant under some rotation $V(\mathbf{R}(\alpha)\mathbf{x}) = V(\mathbf{x})$. Then

$$\langle \mathbf{x} | V U(\alpha) | \psi \rangle = V(\mathbf{x}) \langle \mathbf{x} | U(\alpha) | \psi \rangle = V(\mathbf{x}) \langle \mathbf{R}(\alpha)\mathbf{x} | \psi \rangle, \quad (4.55a)$$

while

$$\langle \mathbf{x} | U(\alpha) V | \psi \rangle = \langle \mathbf{R}(\alpha)\mathbf{x} | V | \psi \rangle = V(\mathbf{R}(\alpha)\mathbf{x}) \langle \mathbf{R}(\alpha)\mathbf{x} | \psi \rangle, \quad (4.55b)$$

and the operator equation $UV = VU$ follows from the equality of $V(\mathbf{R}(\alpha)\mathbf{x})$ and $V(\mathbf{x})$.

In general, finding all the operators that commute with a given Hamiltonian is a very difficult problem. However, it is sometimes possible to deduce conserved quantities by direct inspection. For example, the Hamiltonian for a system of n particles that interact with each other, but not with anything else, is

$$H = \sum_{i=1}^n \frac{\mathbf{p}_i^2}{2m_i} + \sum_{i < j} V(\mathbf{x}_i - \mathbf{x}_j), \quad (4.56)$$

where the potential energy function V only depends on the relative positions of the individual particles. Such a Hamiltonian is invariant under translations of all particles together (shifts of the centre of mass coordinate) and thus the total momentum $\mathbf{p}_{\text{tot}} = \sum_i \mathbf{p}_i$ of this system is conserved.

If the Hamiltonian is a scalar, then $[H, \mathbf{J}] = 0$, $[H, J^2] = 0$ and $[H, P] = 0$ (Problem 4.10), which implies conservation of angular momentum around any axis, conservation of total angular momentum, and conservation of parity. We have already seen that $[\mathbf{J}, J^2] = [\mathbf{J}, P] = 0$, so for a scalar Hamiltonian we can find complete sets of simultaneous eigenkets of H , P , J^2 , and any one of the components of $\mathbf{J}$.

The equation $[H, P] = 0$ implies that if you set up a system that at $t = 0$ is a mirror image of a given system, it will evolve in exactly the same

way as the given system. When the evolution of the mirrored system is watched, it will appear identical to the evolution of the given system when the latter is observed in a mirror. Hence, when $[H, P] = 0$, it is impossible to tell whether a system that is being observed, is being watched directly or through a mirror. One of the major surprises of twentieth-century physics was an experiment by Wu *et al.*⁵ in 1957, which showed that you can see things in a mirror that cannot happen in the real world! That is, the Hamiltonian of the universe is such that $[H, P] \neq 0$.

4.4 The Heisenberg picture

All physical predictions are extracted from the formalism of quantum mechanics by operating with a bra on a ket to extract a complex number: we either calculate the amplitude for some event as $A = \langle \phi | \psi \rangle$ or the expectation value of an observable through $\langle Q \rangle = \langle \psi | Q | \psi \rangle$, where $|Q\rangle \equiv Q|\psi\rangle$. In general our predictions are time-dependent because the state of our system evolves in time according to

$$|\psi, t\rangle = U(t)|\psi, 0\rangle, \quad (4.57)$$

where the time-evolution operator U is defined by equation (4.52).

With every operator of interest we can associate a new time-dependent operator

$$\tilde{Q}_t \equiv U^\dagger(t)QU(t). \quad (4.58)$$

Then at any time t the expectation value of Q can be written

$$\langle Q \rangle_t = \langle \psi, t | Q | \psi, t \rangle = \langle \psi, 0 | U^\dagger(t)QU(t) | \psi, 0 \rangle = \langle \psi, 0 | \tilde{Q}_t | \psi, 0 \rangle. \quad (4.59)$$

That is, the expectation value at time $t = 0$ of the new operator $\tilde{Q}_t$ is equal to the expectation value of the original, physical, operator Q at time t . Similarly, when we wish to calculate an amplitude $\langle \phi, t | \psi, t \rangle$ for something to happen at time t , we can argue that on account of the unitarity of $U(t)$ it is equal to a corresponding amplitude at time zero:

$$\langle \phi, t | \psi, t \rangle = \langle \phi, 0 | \psi, 0 \rangle \quad \text{where} \quad |\phi, t\rangle = U(t)|\phi, 0\rangle. \quad (4.60)$$

Thus if we work with the new time-dependent operators such as $\tilde{Q}_t$, the only states we require are those at $t = 0$. This formalism is called the **Heisenberg picture** to distinguish it from the **Schrödinger picture** in which states evolve and operators are normally time-independent.

As we have seen, classical mechanics applies in the limit that it is sufficient to calculate the expectation values of observables, and is concerned with solving the equations of motion of these expectation values. In the Heisenberg picture quantum mechanics is concerned with solving

⁵ C.S. Wu, E. Ambler, R.W. Hayward, D.D. Hoppes, and R.P. Hudson, *Phys. Rev.*, **105**, 1413 (1957)

the equations of motion of the time-dependent operators $\tilde{Q}_t$, etc. Consequently, there is a degree of similarity between the Heisenberg picture and classical mechanics.

It is straightforward to determine the equation of motion of $\tilde{Q}_t$: we simply differentiate equation (4.58)

$$\frac{d\tilde{Q}_t}{dt} = \frac{dU^\dagger}{dt} Q U + U^\dagger Q \frac{dU}{dt}. \quad (4.61)$$

But differentiating equation (4.52) we have

$$\frac{dU}{dt} = -\frac{iH}{\hbar} U \quad \Rightarrow \quad \frac{dU^\dagger}{dt} = \frac{iH}{\hbar} U^\dagger, \quad (4.62)$$

where we have taken advantage of the fact that U is a function of H and therefore commutes with it. Inserting these expressions into equation (4.61) we obtain

$$\begin{aligned} i\hbar \frac{d\tilde{Q}_t}{dt} &= -HU^\dagger QU + U^\dagger QUH \\ &= [\tilde{Q}_t, H]. \end{aligned} \quad (4.63)$$

This result is similar to Ehrenfest's theorem (eq. 2.34) as it has to be because Ehrenfest's theorem must be recovered if we pre- and post-multiply each side by the time-independent state $|\psi, 0\rangle$.

An extension of the Heisenberg picture is widely used in the quantum theory of fields. In this theory one needs essentially only one state, the vacuum in the remote past $|0\rangle$, which we assume was empty. Excitations of the vacuum are interpreted as particles, each mode of excitation being associated with a different type of particle (photons, electron, up-quarks, etc.). The theory is concerned with the dynamics of operators that excite the vacuum, creating particles, which then propagate to other locations, where they are detected (annihilated) by similar operators. Sometimes one mode of excitation of the vacuum morphs into one or more different modes of the vacuum, and such an event is interpreted as the decay of one type of particle into other particles. The amplitude for any such sequence of events is obtained as a number of the form $\langle 0 | \tilde{A}_1 \tilde{A}_2 \dots \tilde{A}_n | 0 \rangle$, where the operators $\tilde{A}_i$ are creation or annihilation operators for the appropriate particles in the Heisenberg picture.

4.5 What is the essence of quantum mechanics?

It is sometimes said that commutation relations such as $[x_i, p_j] = i\hbar\delta_{ij}$ and $[J_i, J_j] = i\sum_k \epsilon_{ijk} J_k$ are inherently quantum mechanical, but this is not true.

Take for example an ordinary classical rotation matrix $\mathbf{R}(\boldsymbol{\alpha})$ which rotates spatial vectors as $\mathbf{v} \rightarrow \mathbf{v}' = \mathbf{R}(\boldsymbol{\alpha})\mathbf{v}$. Define matrices $\mathcal{J}_x$, $\mathcal{J}_y$ and $\mathcal{J}_z$ via

$$\exp(-i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{J}}) = \mathbf{R}(\boldsymbol{\alpha}), \quad (4.64)$$

where the exponential of a matrix is defined in terms of the power series for e^x . Clearly, the $\mathcal{J}_i$ must be 3×3 matrices, and, since $\mathbf{R}(\boldsymbol{\alpha})$ is real and the angles $\boldsymbol{\alpha}$ are arbitrary, the $\mathcal{J}_i$ must be pure imaginary. Finally, orthogonality of $\mathbf{R}$ requires

$$\begin{aligned} \mathbf{I} &= \mathbf{R}^T(\boldsymbol{\alpha})\mathbf{R}(\boldsymbol{\alpha}) = \exp(-i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{J}})^T \exp(-i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{J}}) \\ &= \exp(-i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{J}}^T) \exp(-i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{J}}). \end{aligned} \quad (4.65)$$

We express $\boldsymbol{\alpha}$ in terms of the angle of the rotation it represents, $\theta = |\boldsymbol{\alpha}|$, and the direction $\mathbf{n} = \boldsymbol{\alpha}/|\boldsymbol{\alpha}|$ of the rotation axis, and then we differentiate equation (4.65) with respect to θ . We obtain

$$\begin{aligned} 0 &= -i\mathbf{n} \cdot \boldsymbol{\mathcal{J}}^T \exp(-i\theta\mathbf{n} \cdot \boldsymbol{\mathcal{J}}^T) \exp(-i\theta\mathbf{n} \cdot \boldsymbol{\mathcal{J}}) \\ &\quad + \exp(-i\theta\mathbf{n} \cdot \boldsymbol{\mathcal{J}}^T) \exp(-i\theta\mathbf{n} \cdot \boldsymbol{\mathcal{J}})(-i\mathbf{n} \cdot \boldsymbol{\mathcal{J}}) \\ &= -i\mathbf{n} \cdot \{\boldsymbol{\mathcal{J}}^T + \boldsymbol{\mathcal{J}}\}. \end{aligned} \quad (4.66)$$

Since $\mathbf{n}$ is an arbitrary unit vector, it now follows that $\mathcal{J}_i^T = -\mathcal{J}_i$, so $\mathcal{J}_i$ is antisymmetric. A pure imaginary antisymmetric matrix is a Hermitian matrix. Thus the $\mathcal{J}_i$ are Hermitian.

For any two vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, it is easy to show that the product $\mathbf{R}^T(\boldsymbol{\alpha})\mathbf{R}(\boldsymbol{\beta})\mathbf{R}(\boldsymbol{\alpha})$ is an orthogonal matrix with determinant +1, so it is a rotation matrix. It leaves the vector $\boldsymbol{\beta}' = \mathbf{R}(-\boldsymbol{\alpha})\boldsymbol{\beta} = \mathbf{R}^T(\boldsymbol{\alpha})\boldsymbol{\beta}$ invariant:

$$\{\mathbf{R}^T(\boldsymbol{\alpha})\mathbf{R}(\boldsymbol{\beta})\mathbf{R}(\boldsymbol{\alpha})\}\boldsymbol{\beta}' = \mathbf{R}^T(\boldsymbol{\alpha})\mathbf{R}(\boldsymbol{\beta})\boldsymbol{\beta} = \mathbf{R}^T(\boldsymbol{\alpha})\boldsymbol{\beta} = \boldsymbol{\beta}'. \quad (4.67)$$

Hence $\boldsymbol{\beta}'$ is the axis of this rotation. Therefore

$$\mathbf{R}^T(\boldsymbol{\alpha})\mathbf{R}(\boldsymbol{\beta})\mathbf{R}(\boldsymbol{\alpha}) = \mathbf{R}(\boldsymbol{\beta}') = \mathbf{R}(\mathbf{R}(-\boldsymbol{\alpha})\boldsymbol{\beta}). \quad (4.68)$$

In Box 4.3 we showed that when $|\boldsymbol{\alpha}|$ is infinitesimal, $\mathbf{R}(-\boldsymbol{\alpha})\boldsymbol{\beta} \simeq \boldsymbol{\beta} - \boldsymbol{\alpha} \times \boldsymbol{\beta}$, so when $\boldsymbol{\beta}$ is also infinitesimal, equation (4.68) can be written in terms of the classical generators (4.64) as

$$(1 + i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{J}})(1 - i\boldsymbol{\beta} \cdot \boldsymbol{\mathcal{J}})(1 - i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{J}}) \simeq 1 - i(\boldsymbol{\beta} - \boldsymbol{\alpha} \times \boldsymbol{\beta}) \cdot \boldsymbol{\mathcal{J}}. \quad (4.69)$$

The zeroth-order terms ('1') and those involving only $\boldsymbol{\alpha}$ or $\boldsymbol{\beta}$ cancel, but the terms involving both $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ cancel only if

$$\alpha_i \beta_j [\mathcal{J}_i, \mathcal{J}_j] = i\alpha_i \beta_j \sum_k \epsilon_{ijk} \mathcal{J}_k. \quad (4.70)$$

This equation can hold for all directions α and β only if the $\mathcal{J}_i$ satisfy

$$[\mathcal{J}_i, \mathcal{J}_j] = i \sum_k \epsilon_{ijk} \mathcal{J}_k, \quad (4.71)$$

which is identical to the ‘quantum’ commutation relation (4.33). Our rederivation of these commutation relations from entirely classical considerations is possible because the relations reflect the fact that the order in which you rotate an object around two different axes matters (Problem 4.6). This is a statement about the geometry of space that has to be recognised by both quantum and classical mechanics.

In Appendix D it is shown that in classical statistical mechanics, each component of position, x_i , and momentum, p_i , is associated with a Hermitian operator $\hat{x}_i$ or $\hat{p}_i$ that acts on functions on phase space. The operator $\hat{p}_i$ generates translations along x_i , while $\hat{x}_i$ generates translations along p_i (boosts). The operators $\hat{L}_i$ associated with angular momentum satisfy the commutation relation $[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$, where $\hbar$ is a number with the same dimensions as $\hbar$ and a magnitude that depends on how $\hat{x}_i$ and $\hat{p}_i$ are normalised.

If the form of the commutation relations is not special to quantum mechanics, what is? In quantum mechanics, complete information about any system is contained in its ket $|\psi\rangle$. There is nothing else. From $|\psi\rangle$ we can evaluate amplitudes such as $\langle \mathbf{x}, \mu | \psi \rangle$ for the system to be found at $\mathbf{x}$ with orientation μ . If we do not care about μ , the total probability for $|\psi\rangle$ to be found at $\mathbf{x}$ is

$$\text{Prob}(\text{at } \mathbf{x} | \psi) = \sum_{\mu} |\langle \mathbf{x}, \mu | \psi \rangle|^2. \quad (4.72)$$

Eigenstates of the $\mathbf{x}$ operator with eigenvalue $\mathbf{x}_0$ are states in which the system is definitely at $\mathbf{x}_0$, while eigenstates of the $\mathbf{p}$ operator with eigenvalue $\hbar \mathbf{k}$ are states in which the system definitely has momentum $\hbar \mathbf{k}$.

By contrast, in classical statistical mechanics we declare at the outset that a well-defined state is one that has definite values for all measurable quantities, so it has a definite position, momentum, orientation, etc. The eigenfunctions of $\hat{\mathbf{p}}$ or $\hat{\mathbf{L}}$ do not represent states of definite momentum or angular momentum, because we have already defined what such states are.

Classical statistical mechanics knows nothing about probability amplitudes, but interprets the functions on phase space on which $\hat{\mathbf{p}}$ or $\hat{\mathbf{L}}$ act as probability distributions. This is possible because, as we show in Appendix D, the integral of such a distribution can be normalised to one and is conserved. We can certainly expand any such distribution in the eigenfunctions of, say $\hat{\mathbf{p}}$. However, as in quantum mechanics the expansion coefficients will not be positive – in fact, they will generally be complex. Hence they cannot be interpreted as probabilities. What makes quantum mechanics fundamentally different is its reliance on complex quantum amplitudes, and the physical interpretation that it gives to a functional

expansion through the fundamental rule (1.4) for adding quantum amplitudes. Quantum mechanics is therefore naturally formulated in terms of states $|\psi\rangle$ that inhabit a complex vector space of arbitrary dimension – a so-called **Hilbert space**. These states may always be expanded in terms of a complete set of eigenstates of a Hermitian operator, and the (complex) expansion coefficients have a simple physical interpretation.

Classical statistical mechanics is restricted to probabilities, which have to be real, non-negative numbers and are therefore never expansion coefficients. Quantum and classical mechanics incorporate the same commutation relations, however, because, as we stressed in §4.2, these follow from the geometry of space. From a mathematician's perspective, the commutation relations of quantum-mechanical operators and the operators of classical statistical physics have to be the same because both systems of operators provide representations of the 'Lie algebra' of the same mathematical group (Appendix E).

Problems

4.1 Verify that $[\mathbf{J}, \mathbf{x} \cdot \mathbf{x}] = 0$ and $[\mathbf{J}, \mathbf{x} \cdot \mathbf{p}] = 0$ by using the commutation relations $[x_i, J_j] = i \sum_k \epsilon_{ijk} x_k$ and $[p_i, J_j] = i \sum_k \epsilon_{ijk} p_k$.

4.2* Show that the vector product $\mathbf{a} \times \mathbf{b}$ of two classical vectors transforms like a vector under rotations. Hint: A rotation matrix $\mathbf{R}$ satisfies the relations $\mathbf{R} \cdot \mathbf{R}^T = \mathbf{I}$ and $\det(\mathbf{R}) = 1$, which in tensor notation read $\sum_p R_{ip} R_{tp} = \delta_{it}$ and $\sum_{ijk} \epsilon_{ijk} R_{ir} R_{js} R_{kt} = \epsilon_{rst}$.

4.3* We have shown that $[v_i, J_j] = i \sum_k \epsilon_{ijk} v_k$ for any operator whose components v_i form a vector. The expectation value of this operator relation in any state $|\psi\rangle$ is then $\langle \psi | [v_i, J_j] | \psi \rangle = i \sum_k \epsilon_{ijk} \langle \psi | v_k | \psi \rangle$. Check that with $U(\boldsymbol{\alpha}) = e^{-i\boldsymbol{\alpha} \cdot \mathbf{J}}$ this relation is consistent under a further rotation $|\psi\rangle \rightarrow |\psi'\rangle = U(\boldsymbol{\alpha})|\psi\rangle$ by evaluating both sides separately.

4.4* The matrix for rotating an ordinary vector by ϕ around the z -axis is

$$\mathbf{R}(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4.73)$$

By considering the form taken by $\mathbf{R}$ for infinitesimal ϕ calculate from $\mathbf{R}$ the matrix $\mathbf{J}_z$ that appears in $\mathbf{R}(\phi) = \exp(-i\mathbf{J}_z\phi)$. Introduce new coordinates $u_1 = (-x + iy)/\sqrt{2}$, $u_2 = z$ and $u_3 = (x + iy)/\sqrt{2}$. Write down the matrix $\mathbf{M}$ that appears in $\mathbf{u} = \mathbf{M} \cdot \mathbf{x}$ [where $\mathbf{x} \equiv (x, y, z)$] and show that it is unitary. Then show that

$$\mathbf{J}_z' = \mathbf{M} \cdot \mathbf{J}_z \cdot \mathbf{M}^\dagger \quad (4.74)$$

is identical with S_z in the set of spin-one Pauli analogues

$$S_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad S_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (4.75)$$

Write down the matrix $\mathcal{J}_x$ whose exponential generates rotations around the x -axis, calculate $\mathcal{J}'_x$ by analogy with equation (4.74) and check that your result agrees with S_x in the set (4.75). Explain as fully as you can the meaning of these calculations.

4.5 Determine the commutator $[\mathcal{J}'_x, \mathcal{J}'_z]$ of the generators used in Problem 4.4. Show that it is equal to $-i\mathcal{J}'_y$, where $\mathcal{J}'_y$ is identical with S_y in the set (4.75).

4.6* Show that if α and β are non-parallel vectors, α is not invariant under the combined rotation $\mathbf{R}(\alpha)\mathbf{R}(\beta)$. Hence show that

$$\mathbf{R}^T(\beta)\mathbf{R}^T(\alpha)\mathbf{R}(\beta)\mathbf{R}(\alpha)$$

is not the identity operation. Explain the physical significance of this result.

4.7* In this problem you derive the wavefunction

$$\langle \mathbf{x} | \mathbf{p} \rangle = e^{i\mathbf{p} \cdot \mathbf{x} / \hbar} \quad (4.76)$$

of a state of well-defined momentum from the properties of the translation operator $U(\mathbf{a})$. The state $|\mathbf{k}\rangle$ is one of well-defined momentum $\hbar\mathbf{k}$. How would you characterise the state $|\mathbf{k}'\rangle = U(\mathbf{a})|\mathbf{k}\rangle$? Show that the wavefunctions of these states are related by $u_{\mathbf{k}'}(\mathbf{x}) = e^{-i\mathbf{a} \cdot \mathbf{k}} u_{\mathbf{k}}(\mathbf{x})$ and $u_{\mathbf{k}'}(\mathbf{x}) = u_{\mathbf{k}}(\mathbf{x} - \mathbf{a})$. Hence obtain equation (4.76).

4.8 By expanding the anticommutator on the left and then applying the third rule of the set (2.22), show that any three operators satisfy the identity

$$[\{A, B\}, C] = \{A, [B, C]\} + \{[A, C], B\}. \quad (4.77)$$

4.9 Define G in terms of the parity operator P by

$$G = \frac{1}{2}(1 - P). \quad (4.78)$$

Show that G is Hermitian and that $G^n = G$ for positive integer n . Explain this result in terms of the eigenkets and eigenvalues of G . Show further that $P = U(\pi)$ where $U(s) = e^{isG}$.

4.10 Let P be the parity operator and S an arbitrary scalar operator. Explain why P and S must commute.

4.11 In this problem we consider discrete transformations other than that associated with parity. Let $\mathcal{S}$ be a linear transformation on ordinary three-dimensional space that effects a reflection in a plane. Let S be the associated operator on kets. Explain the physical relationship between the kets $|\psi\rangle$ and $|\psi'\rangle = S|\psi\rangle$. Explain why we can write

$$S\langle\psi|\mathbf{x}|\psi\rangle = \langle\psi|S^\dagger\mathbf{x}S|\psi\rangle. \quad (4.79)$$

What are the possible eigenvalues of S ?

Given that S reflects in the plane through the origin with unit normal $\hat{\mathbf{n}}$, show, by means of a diagram or otherwise, that its matrix is given by

$$S_{ij} = \delta_{ij} - 2n_i n_j. \quad (4.80)$$

Determine the form of this matrix in the case that $\mathbf{n} = (1, -1, 0)/\sqrt{2}$. Show that in this case $Sx = yS$ and give an alternative expression for Sy .

Show that a potential of the form

$$V(\mathbf{x}) = f(R) + \lambda xy, \quad \text{where } R = \sqrt{x^2 + y^2} \quad (4.81)$$

satisfies $V(S\mathbf{x}) = V(\mathbf{x})$ and explain the geometrical significance of this equation. Show that $[S, V] = 0$. Given that E is an eigenvalue of $H = p^2/2m + V$ that has a unique eigenket $|E\rangle$, what equation does $|E\rangle$ satisfy in addition to $H|E\rangle = E|E\rangle$?

4.12 Show that the operator defined by $\langle x, y|S|\psi\rangle = \langle y, x|\psi\rangle$ is Hermitian.